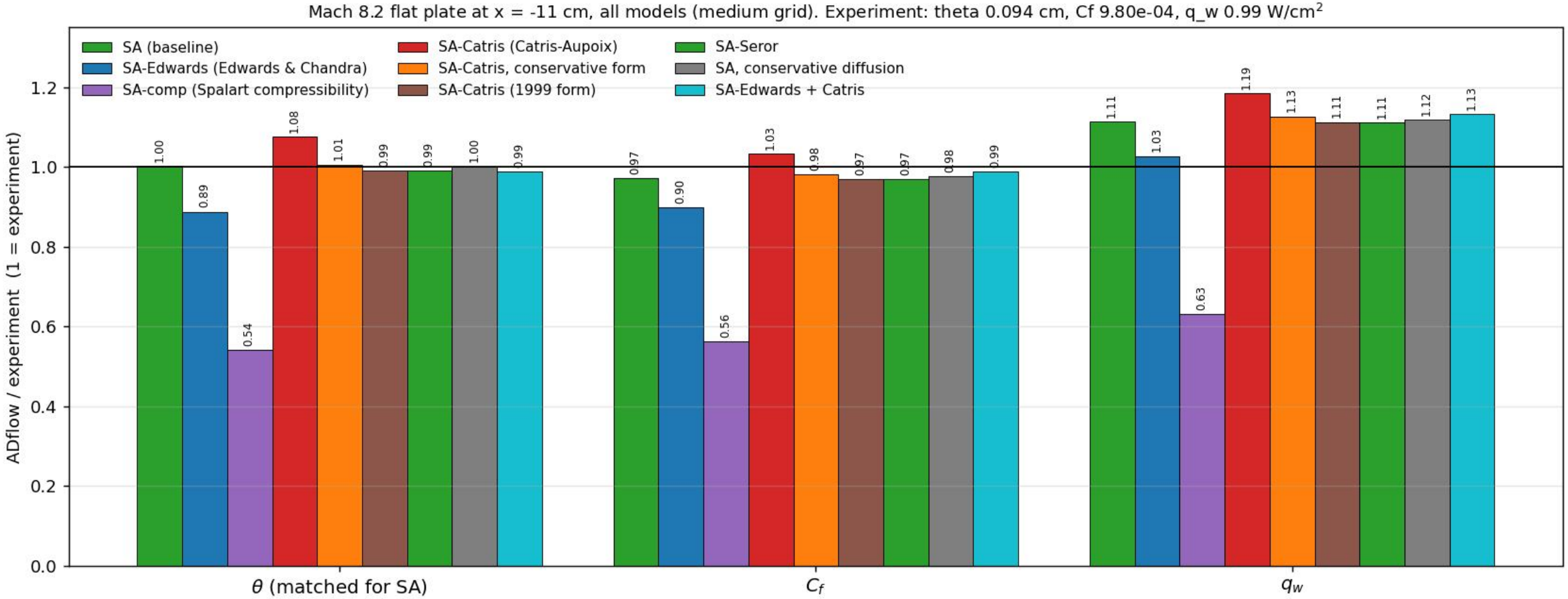
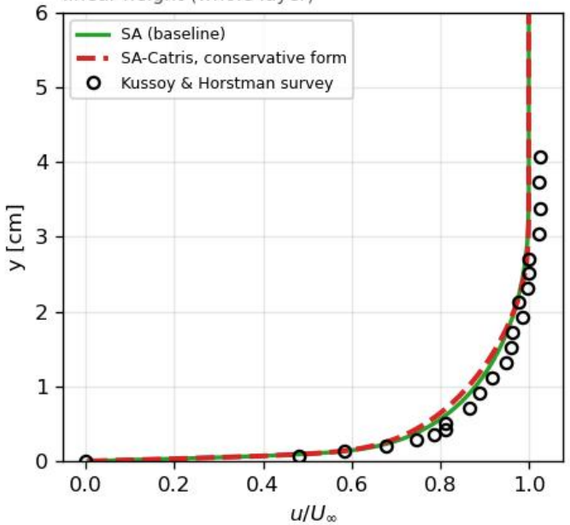
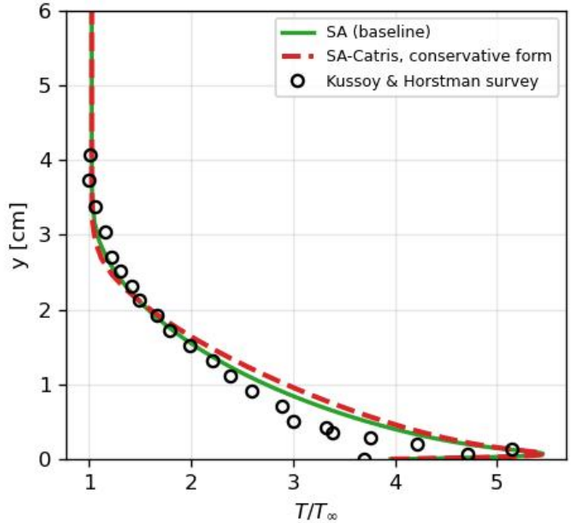
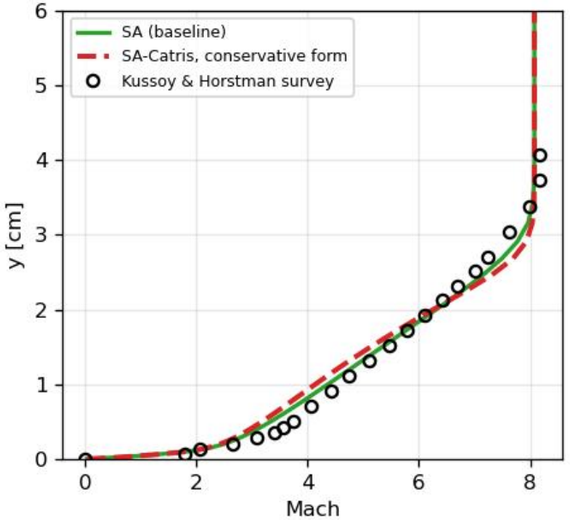

Mach 8.2 flat plate at x = -11 cm, all models (medium grid). Experiment: theta 0.094 cm, Cf 9.80e-04, q_w 0.99 W/cm²
SA (baseline)
SA-Edwards (Edwards & Chandra)
SA-comp (Spalart compressibility)
SA-Catris (Catris-Aupoix)
SA-Catris, conservative form
SA-Catris (1999 form)
SA-Seror
SA, conservative diffusion
SA-Edwards + Catris
ADflow / experiment (1 = experiment)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.00
0.89
0.54
1.08
1.01
0.99
0.99
1.00
0.99
0.97
0.90
0.56
1.03
0.98
0.97
0.97
0.98
0.99
1.11
1.03
0.63
1.19
1.13
1.11
1.11
1.12
1.13
θ (matched for SA)
C_f
q_w

Mach 8.2 flat plate, incoming boundary layer at x = -11 cm: SA-Catris, conservative form vs SA vs experiment
Catris-Aupoix density correction, conservative left-hand side (TMR SA-noft2-CatrisCons)

linear height (whole layer)



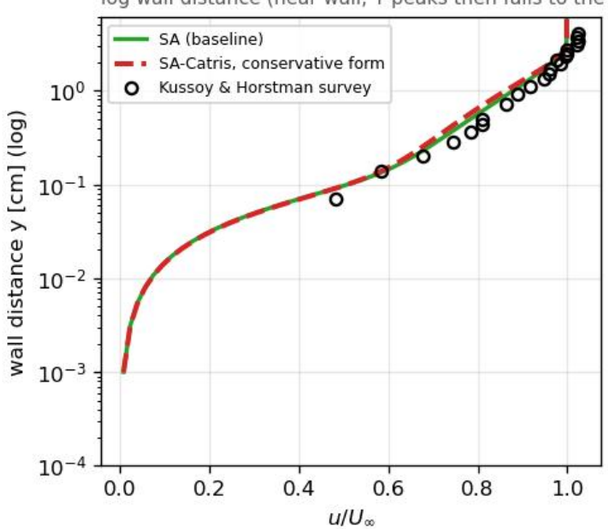


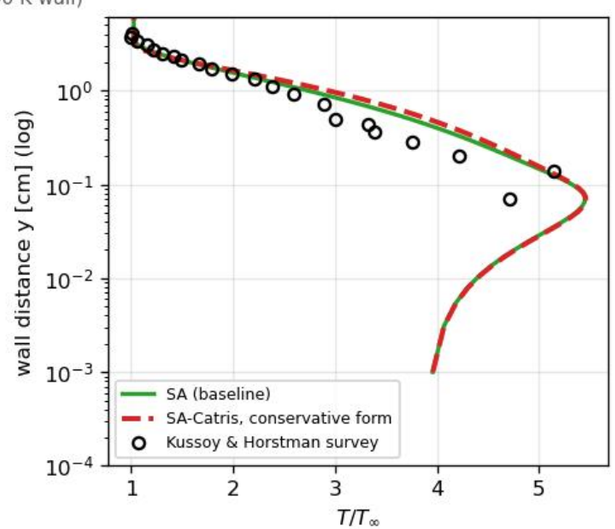


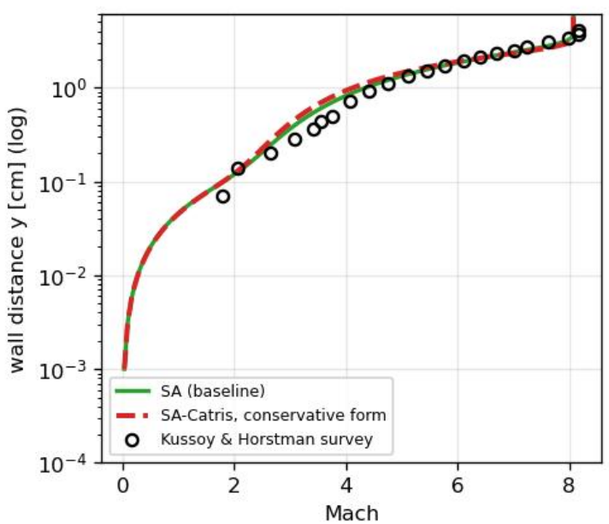
x = -11 cm; model = SA-Catris-cons
(theta matches for SA by construction)

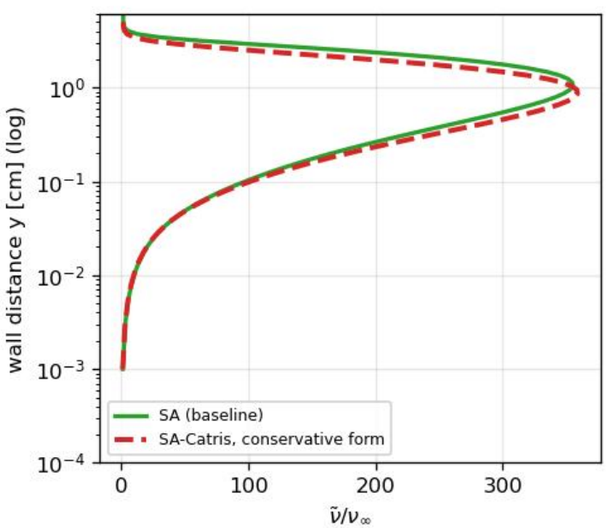
quantity	exp.	SA	model	model vs SA	model vs exp.
theta, cm	0.0940	0.0942	0.0946	+0.4%	+0.6%
d99, cm	-	2.635	2.523	-4.3%	-
Cf x 1e3	0.980	0.953	0.962	+1.0%	-1.8%
q_w, W/cm2	0.990	1.102	1.116	+1.2%	+12.7%

log wall distance (near wall; T peaks then falls to the 300 K wall)



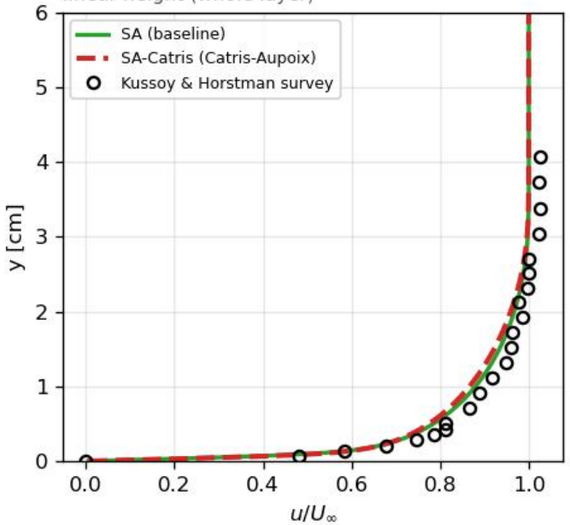


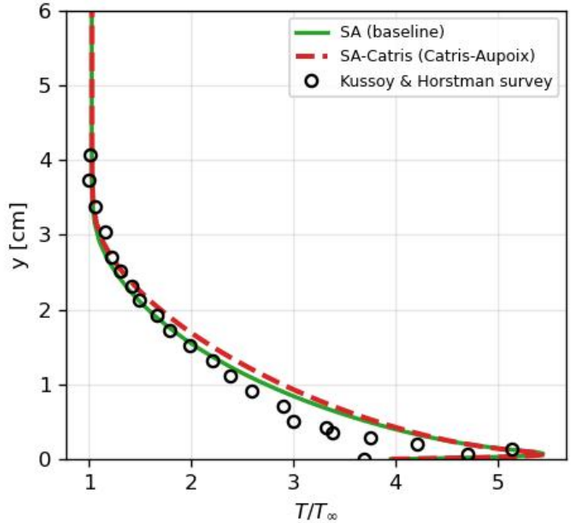


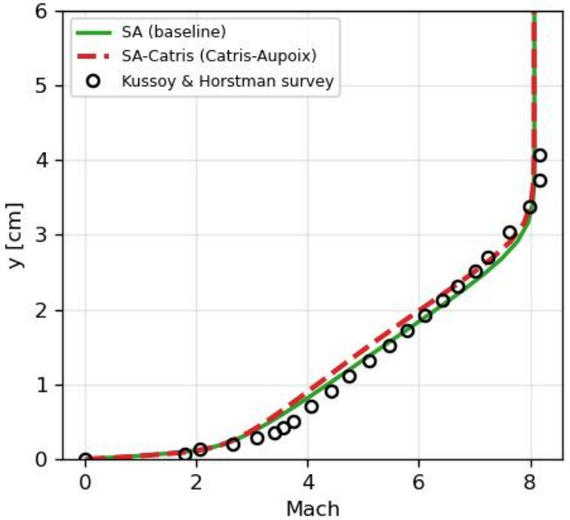


Mach 8.2 flat plate, incoming boundary layer at x = -11 cm: SA-Catris (Catris-Aupoix) vs SA vs experiment
Catris-Aupoix density correction (2000): diffuses sqrt(rho) nuTilde, laminar diffusion of mu, original non-conservative left-hand side

linear height (whole layer)



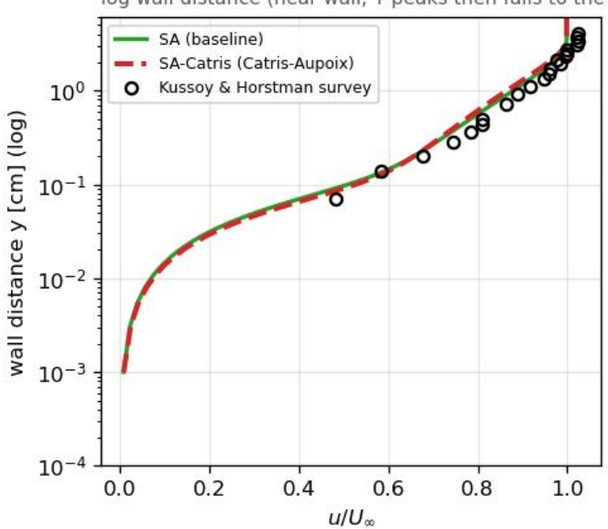


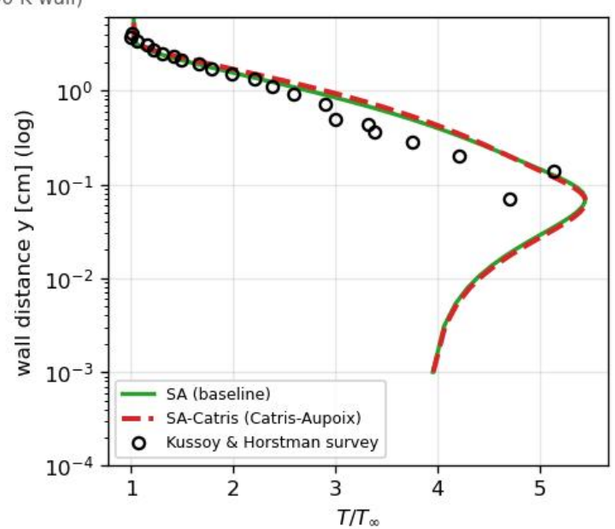


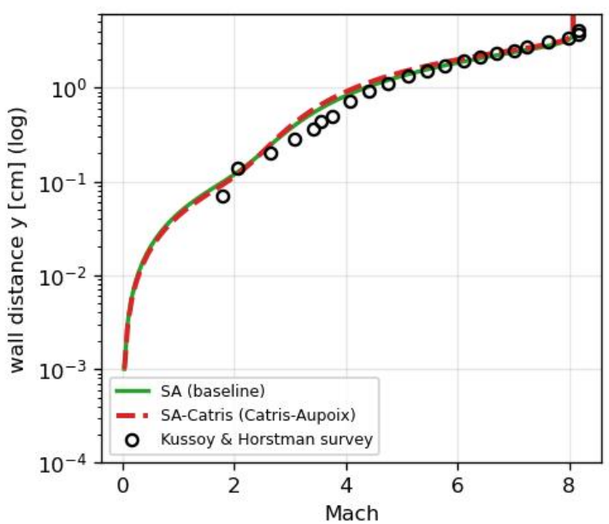
x = -11 cm; model = SA-Catris
(theta matches for SA by construction)

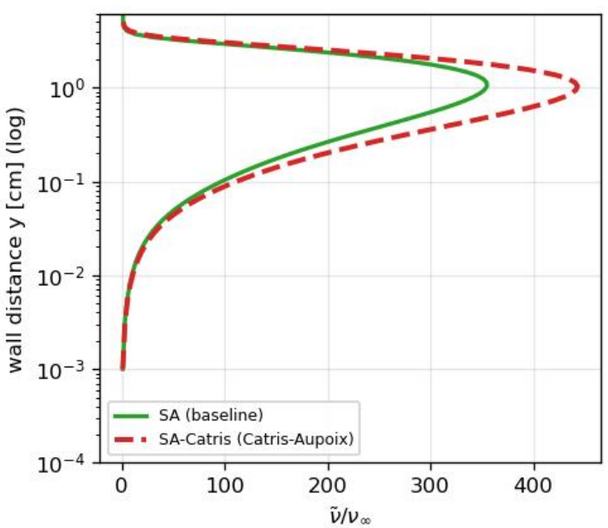
quantity	exp.	SA	model	model vs SA	model vs exp.
theta, cm	0.0940	0.0942	0.1011	+7.4%	+7.6%
d99, cm	-	2.635	2.757	+4.6%	-
Cf x 1e3	0.980	0.953	1.013	+6.3%	+3.4%
q_w, W/cm2	0.990	1.102	1.173	+6.5%	+18.5%

log wall distance (near wall; T peaks then falls to the 300 K wall)



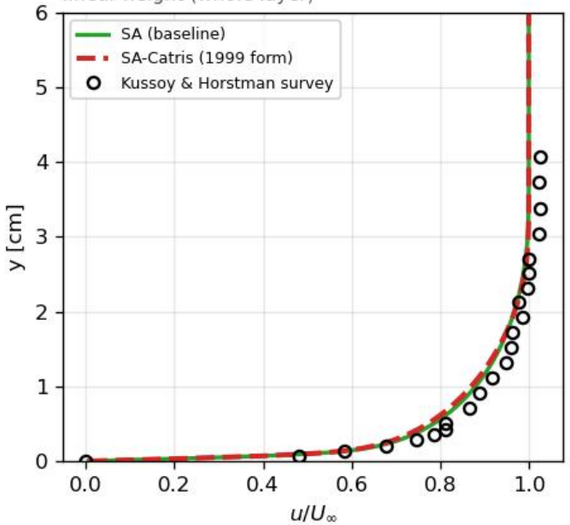


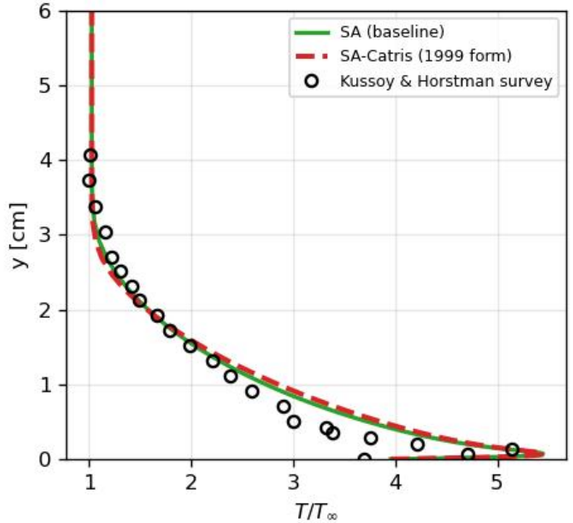


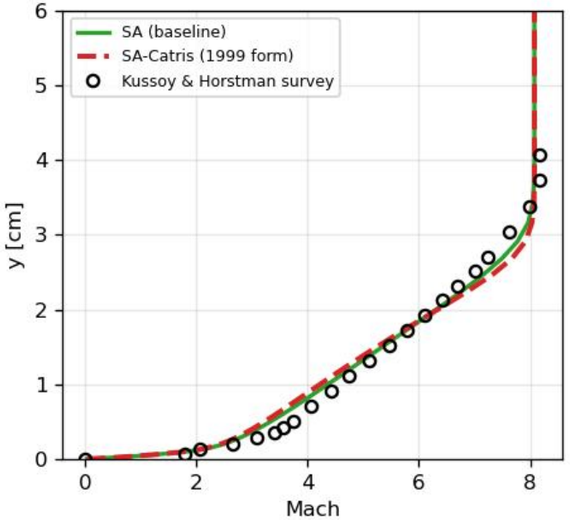


Mach 8.2 flat plate, incoming boundary layer at x = -11 cm: SA-Catris (1999 form) vs SA vs experiment
Catris (1999) form of the density-gradient diffusion terms

linear height (whole layer)



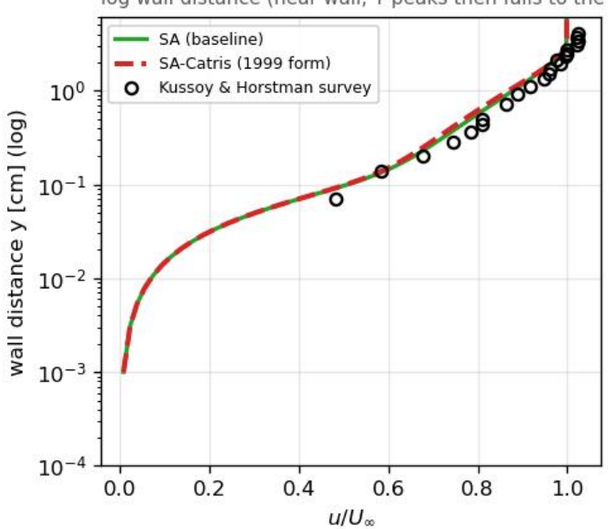


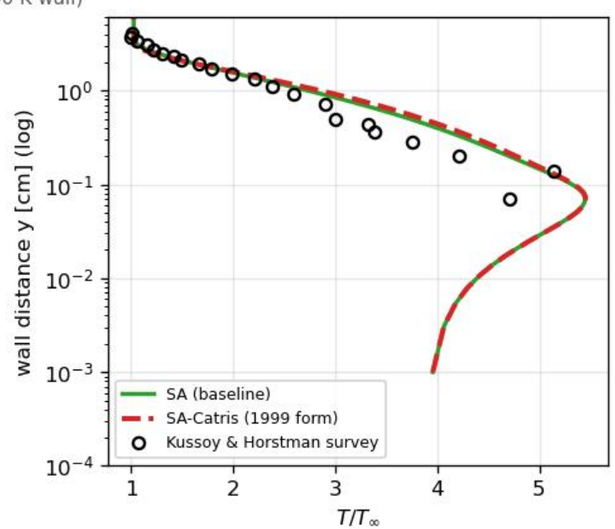


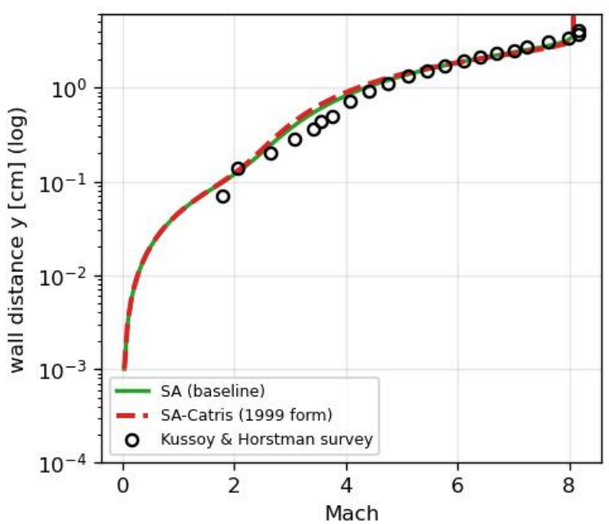
x = -11 cm; model = SA-Catris1999
(theta matches for SA by construction)

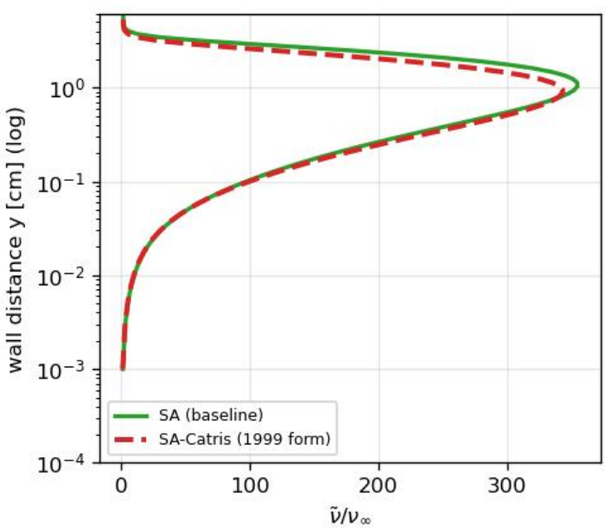
quantity	exp.	SA	model	model vs SA	model vs exp.
theta, cm	0.0940	0.0942	0.0932	-1.0%	-0.8%
d99, cm	-	2.635	2.526	-4.2%	-
Cf x 1e3	0.980	0.953	0.951	-0.2%	-3.0%
q_w, W/cm2	0.990	1.102	1.102	-0.0%	+11.3%

log wall distance (near wall; T peaks then falls to the 300 K wall)



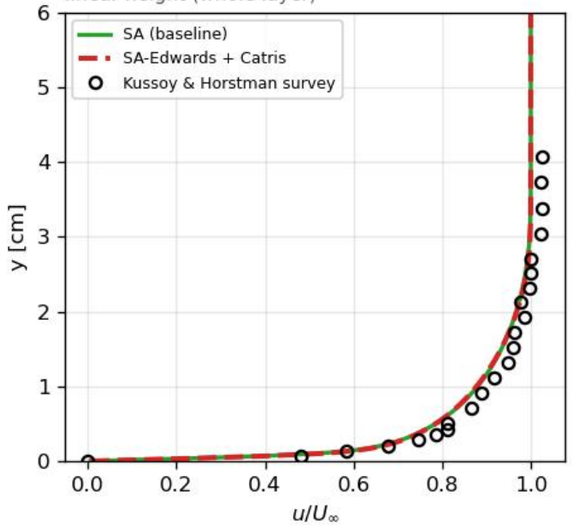


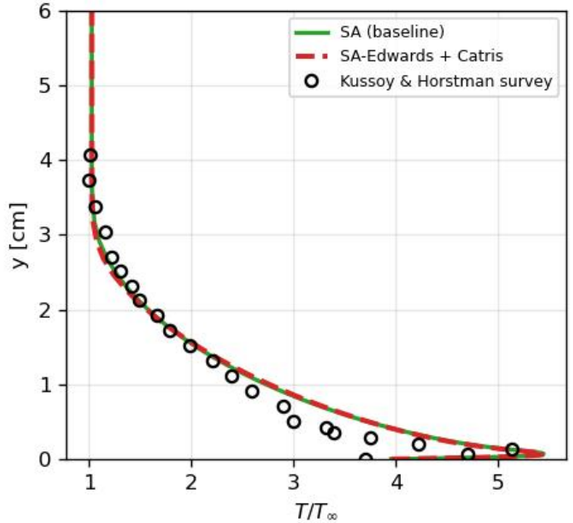


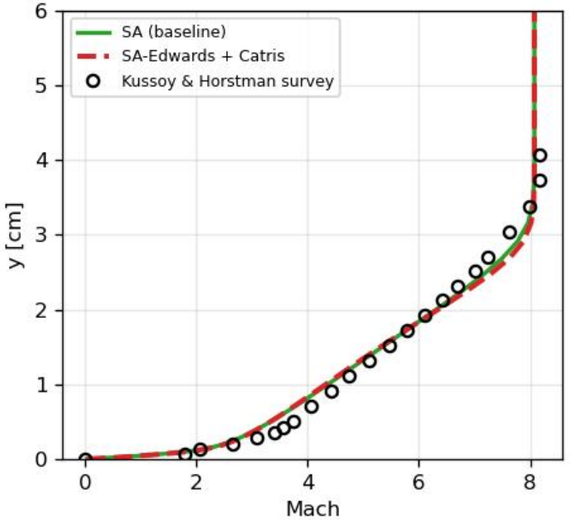


Mach 8.2 flat plate, incoming boundary layer at x = -11 cm: SA-Edwards + Catris vs SA vs experiment
SA-Edwards combined with the Catris-Aupoix density correction

linear height (whole layer)



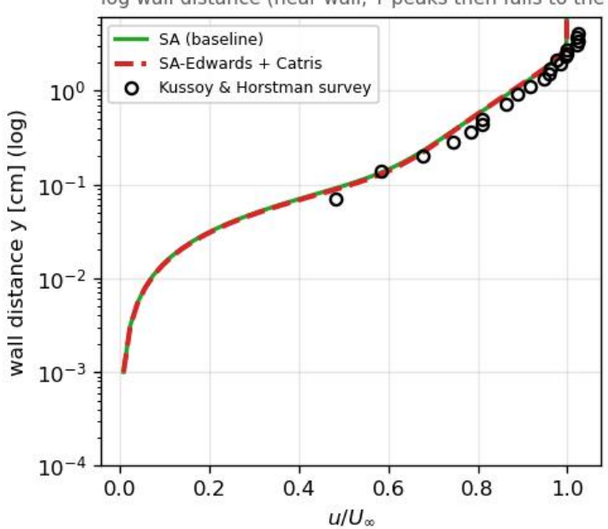


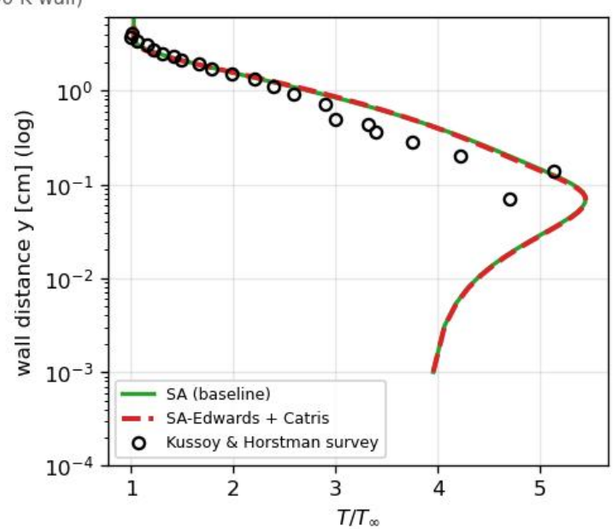


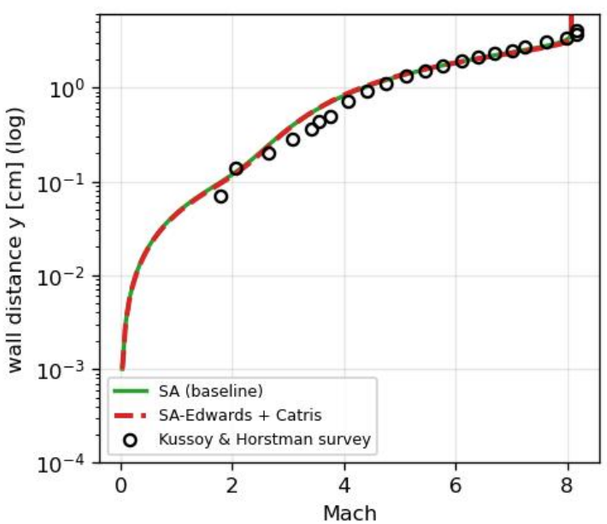
x = -11 cm; model = SA-Edwards-Catris
(theta matches for SA by construction)

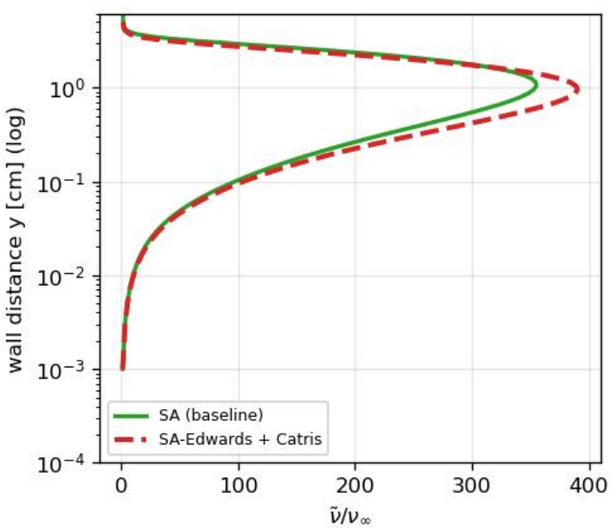
quantity	exp.	SA	model	model vs SA	model vs exp.
theta, cm	0.0940	0.0942	0.0928	-1.4%	-1.2%
d99, cm	-	2.635	2.555	-3.0%	-
Cf x 1e3	0.980	0.953	0.969	+1.7%	-1.1%
q_w, W/cm2	0.990	1.102	1.122	+1.8%	+13.3%

log wall distance (near wall; T peaks then falls to the 300 K wall)



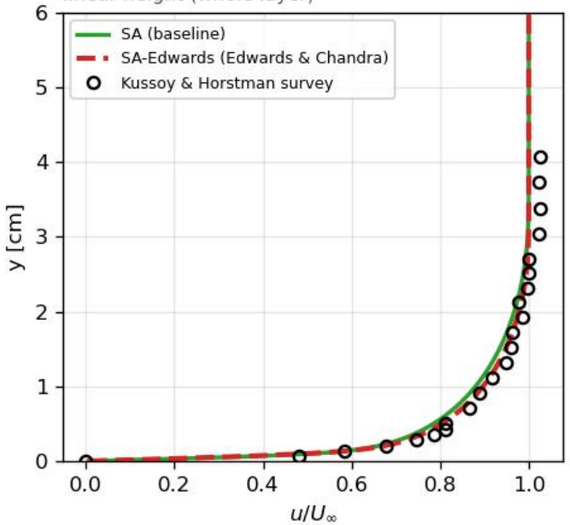


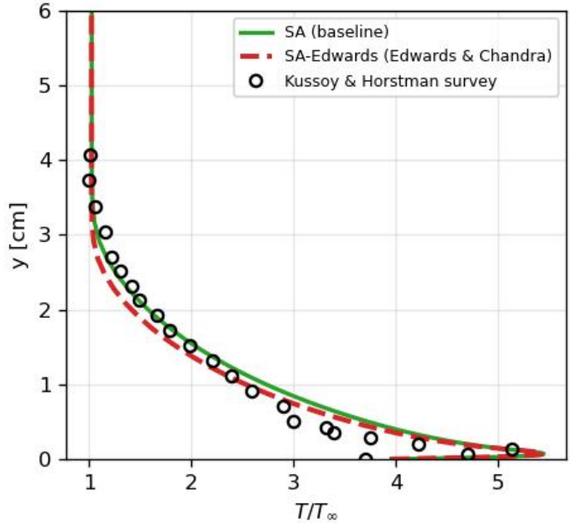


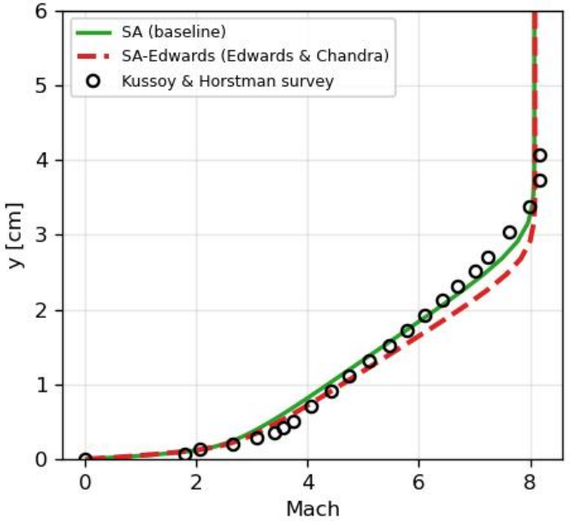


Mach 8.2 flat plate, incoming boundary layer at x = -11 cm: SA-Edwards (Edwards & Chandra) vs SA vs experiment
Edwards & Chandra (AIAA 94-2275): strain-based production, Shat = S (1/chi + fv1), r = tanh(nuTilde/(Shat k^2 d^2))/tanh(1), no ft2

linear height (whole layer)



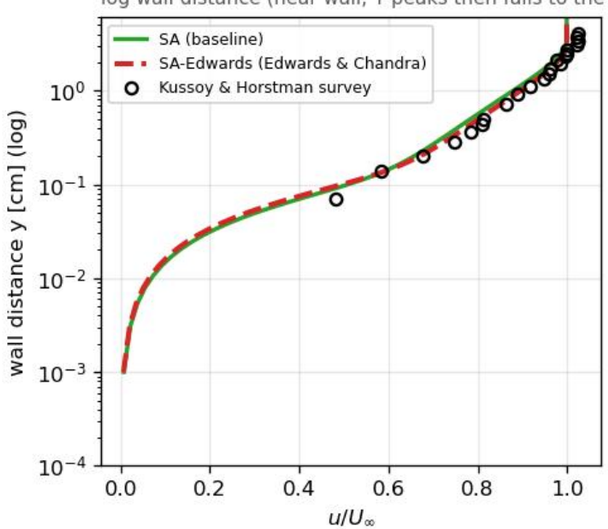


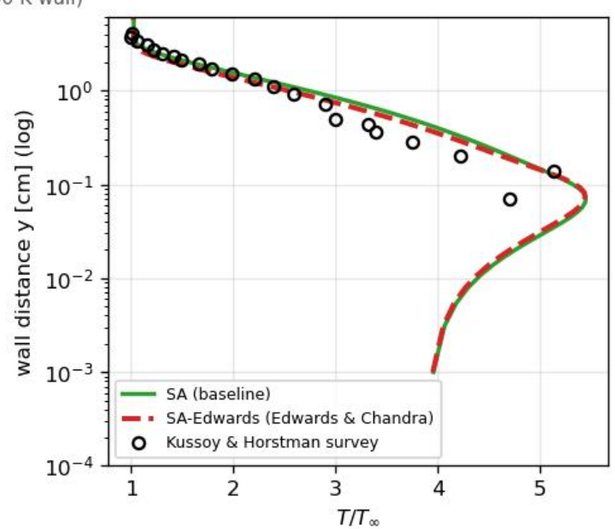


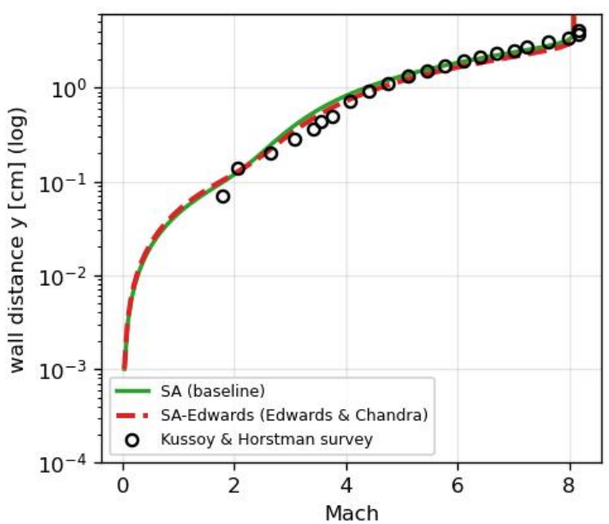
x = -11 cm; model = SA-Edwards
(theta matches for SA by construction)

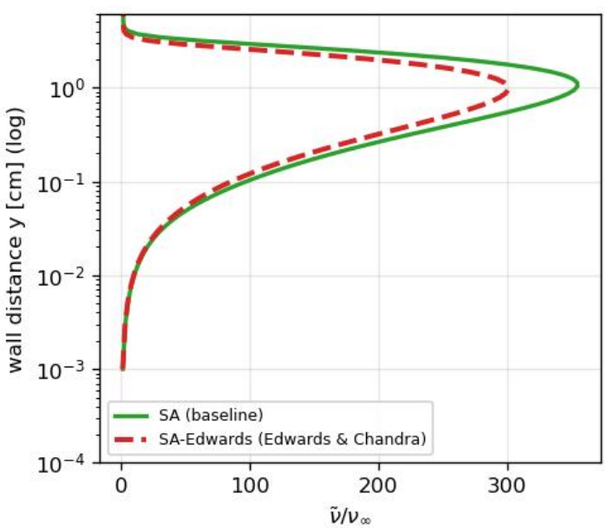
quantity	exp.	SA	model	model vs SA	model vs exp.
theta, cm	0.0940	0.0942	0.0835	-11.4%	-11.2%
d99, cm	-	2.635	2.376	-9.8%	-
Cf x 1e3	0.980	0.953	0.881	-7.5%	-10.1%
q_w, W/cm2	0.990	1.102	1.017	-7.7%	+2.7%

log wall distance (near wall; T peaks then falls to the 300 K wall)



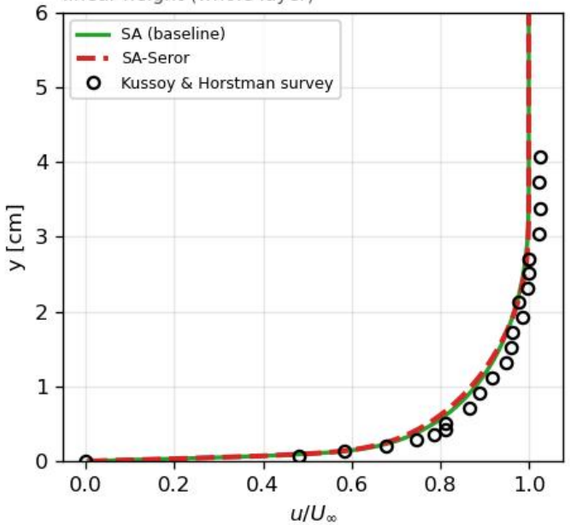


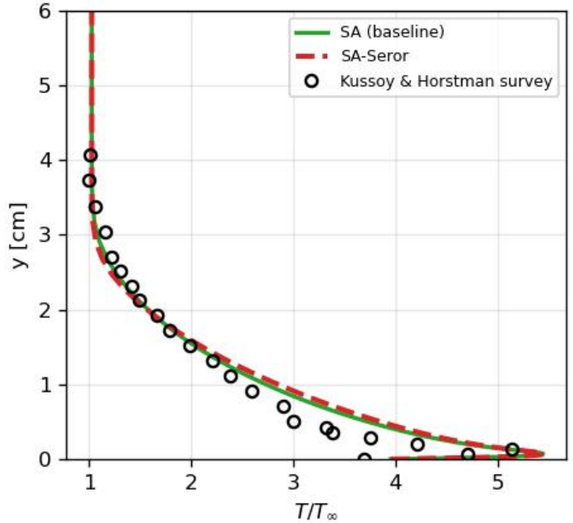


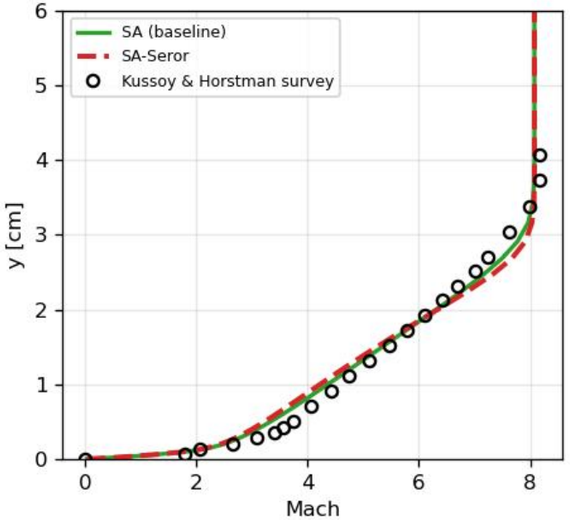


Mach 8.2 flat plate, incoming boundary layer at x = -11 cm: SA-Seror vs SA vs experiment
Seror et al. (2005) form of the density-gradient diffusion terms

linear height (whole layer)



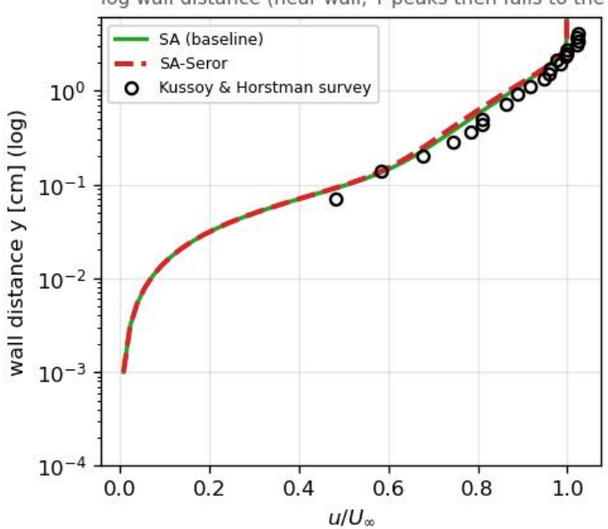


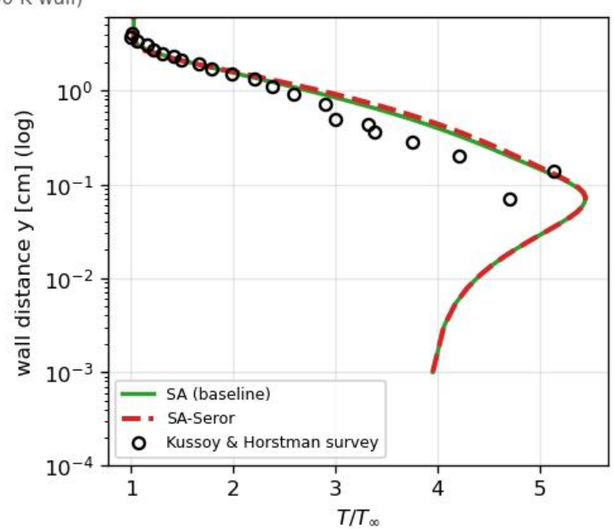


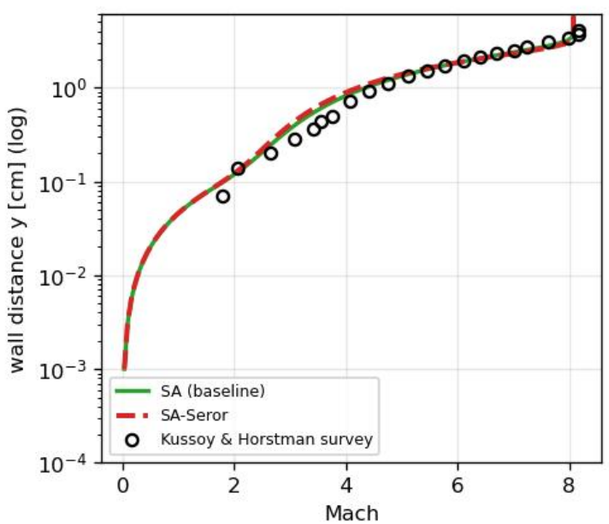
x = -11 cm; model = SA-Seror
(theta matches for SA by construction)

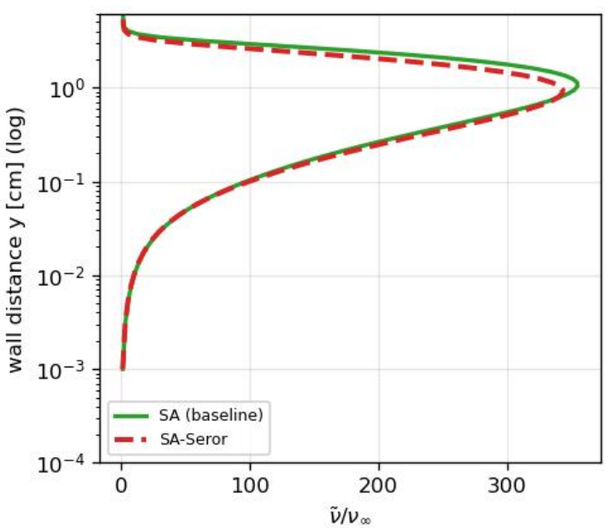
quantity	exp.	SA	model	model vs SA	model vs exp.
theta, cm	0.0940	0.0942	0.0932	-1.0%	-0.8%
d99, cm	-	2.635	2.527	-4.1%	-
Cf x 1e3	0.980	0.953	0.950	-0.3%	-3.0%
q_w, W/cm2	0.990	1.102	1.101	-0.1%	+11.2%

log wall distance (near wall; T peaks then falls to the 300 K wall)



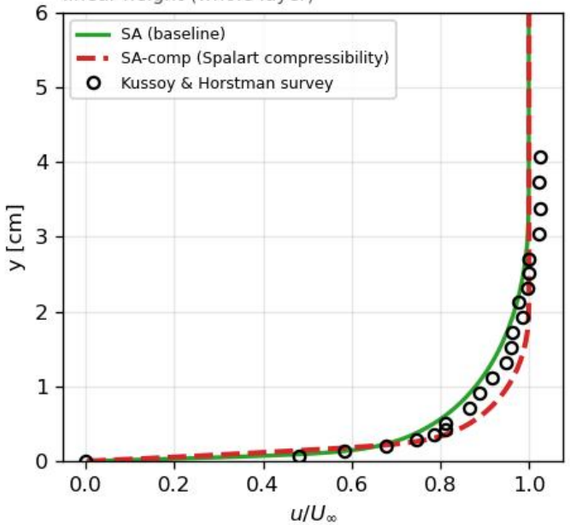


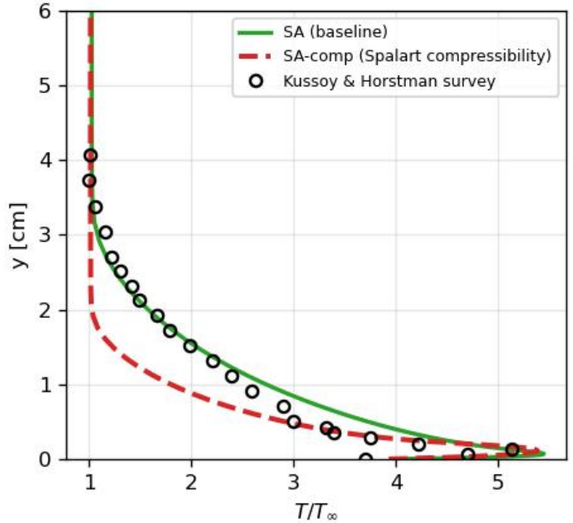


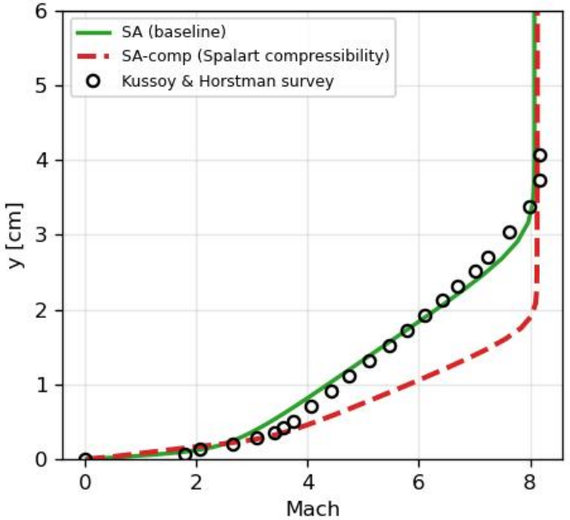


Mach 8.2 flat plate, incoming boundary layer at x = -11 cm: SA-comp (Spalart compressibility) vs SA vs experiment
Spalart compressibility correction (AIAA 2000-2306): extra destruction -C5 nuTilde^2 (du_i/dx_j)^2 / a^2, C5 = 3.5

linear height (whole layer)



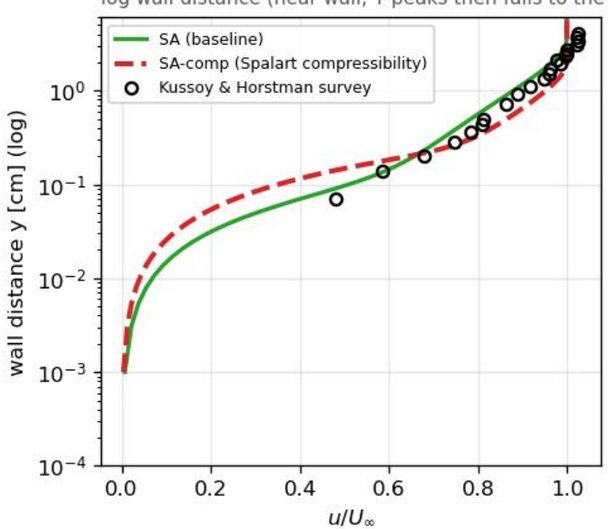


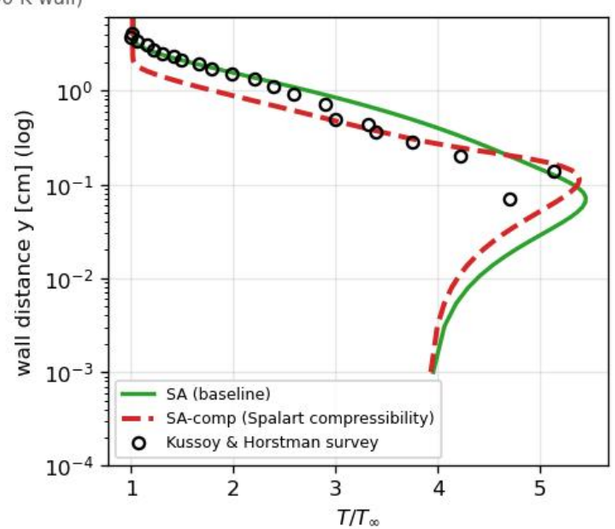


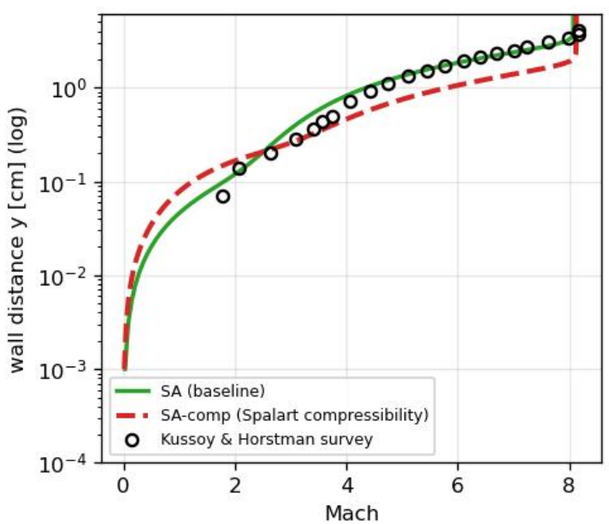
x = -11 cm; model = SA-comp
(theta matches for SA by construction)

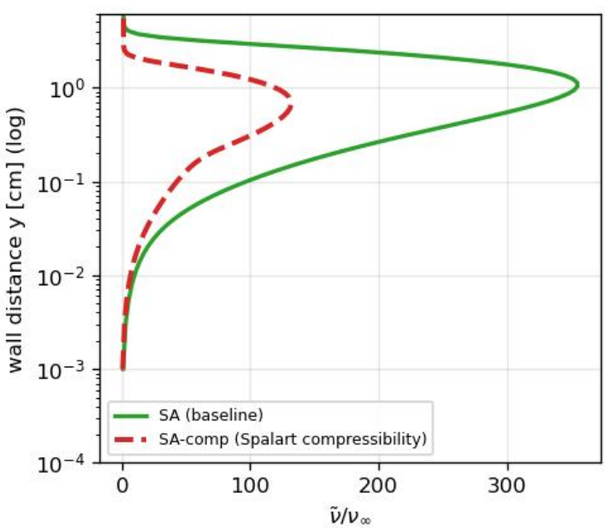
quantity	exp.	SA	model	model vs SA	model vs exp.
theta, cm	0.0940	0.0942	0.0508	-46.0%	-45.9%
d99, cm	-	2.635	1.548	-41.3%	-
Cf x 1e3	0.980	0.953	0.551	-42.2%	-43.8%
q_w, W/cm2	0.990	1.102	0.625	-43.3%	-36.9%

log wall distance (near wall; T peaks then falls to the 300 K wall)



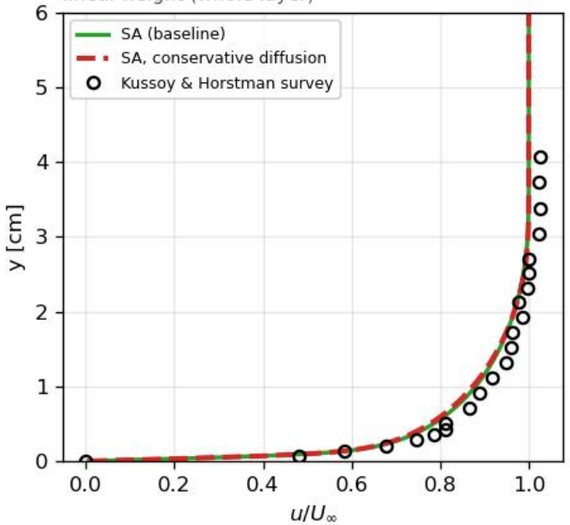


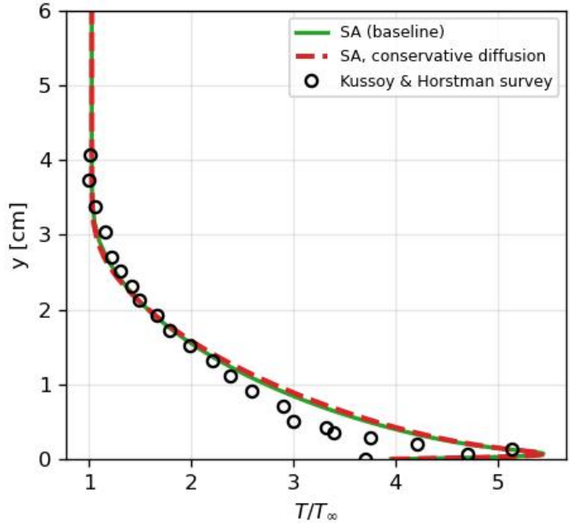


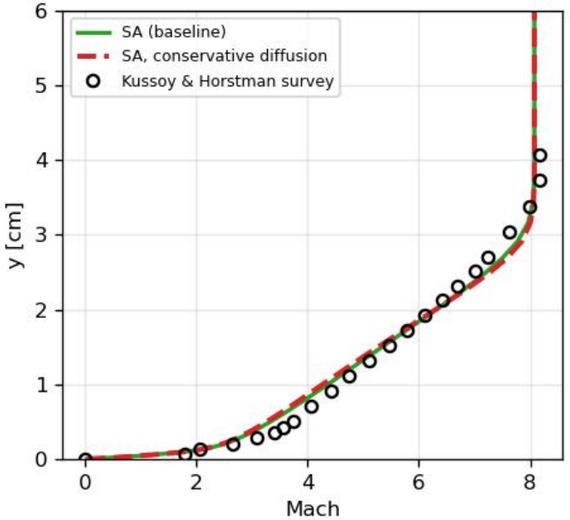


Mach 8.2 flat plate, incoming boundary layer at x = -11 cm: SA, conservative diffusion vs SA vs experiment
conservative diffusion form (TMR / SU2): adds the density-gradient cross terms

linear height (whole layer)



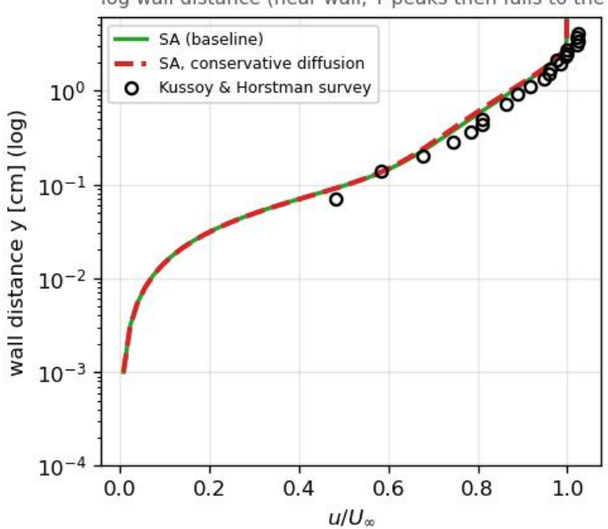


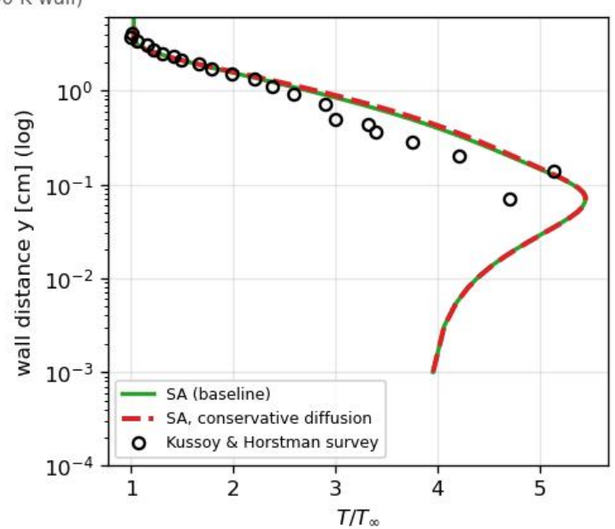


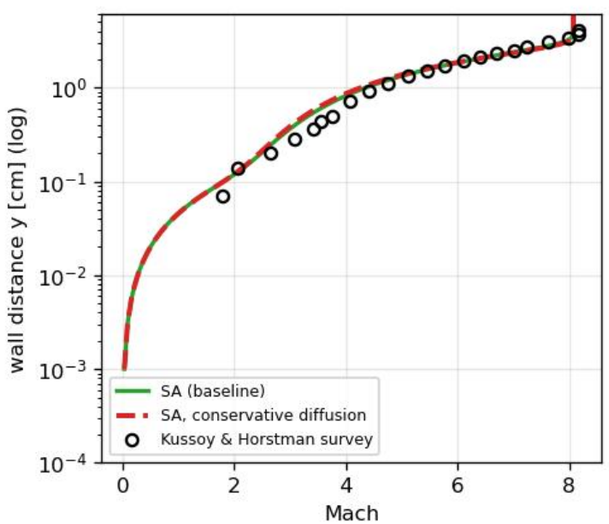
x = -11 cm; model = SA-cons
(theta matches for SA by construction)

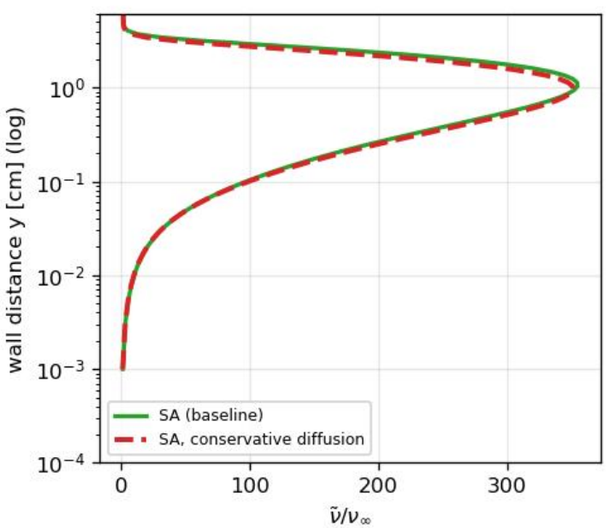
quantity	exp.	SA	model	model vs SA	model vs exp.
theta, cm	0.0940	0.0942	0.0941	-0.1%	+0.1%
d99, cm	-	2.635	2.586	-1.9%	-
Cf x 1e3	0.980	0.953	0.956	+0.4%	-2.4%
q_w, W/cm2	0.990	1.102	1.108	+0.5%	+11.9%

log wall distance (near wall; T peaks then falls to the 300 K wall)

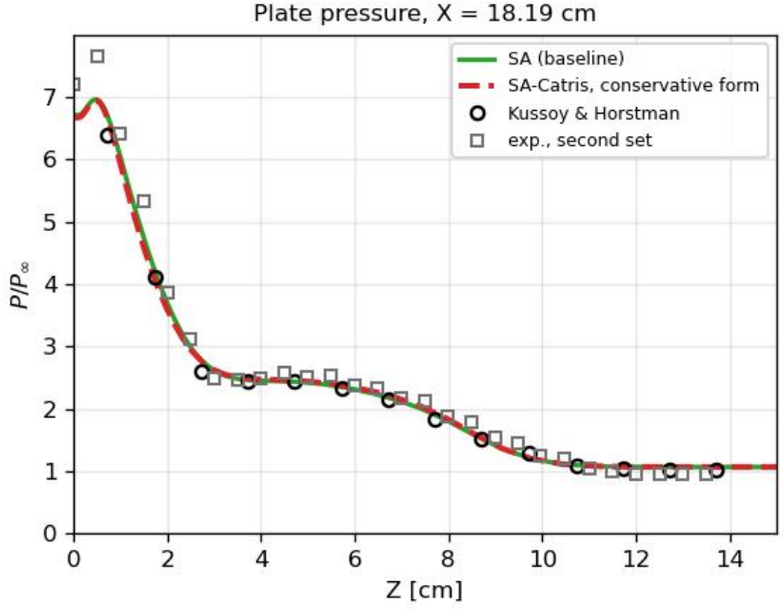


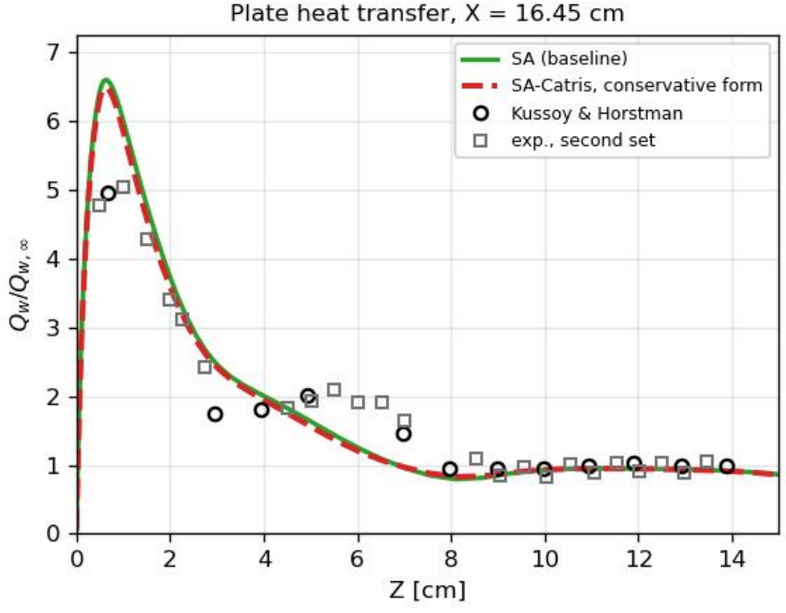


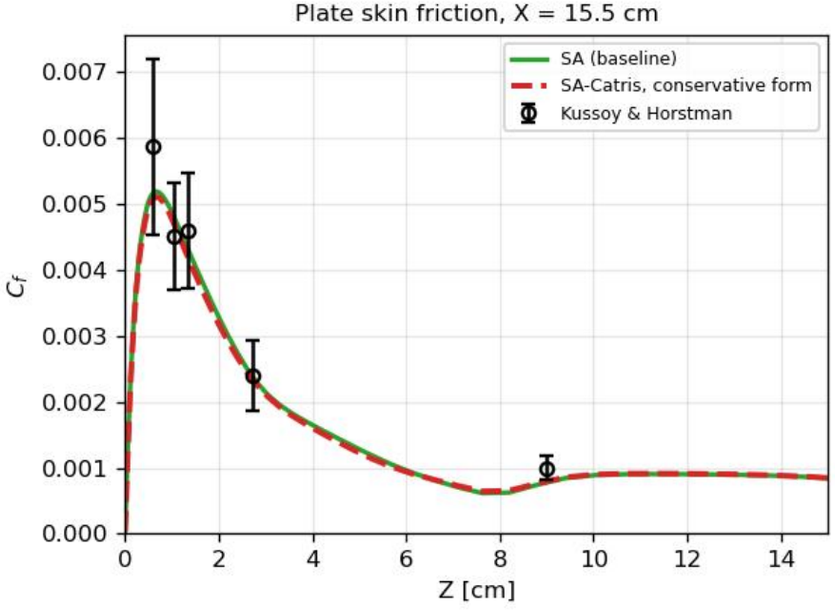


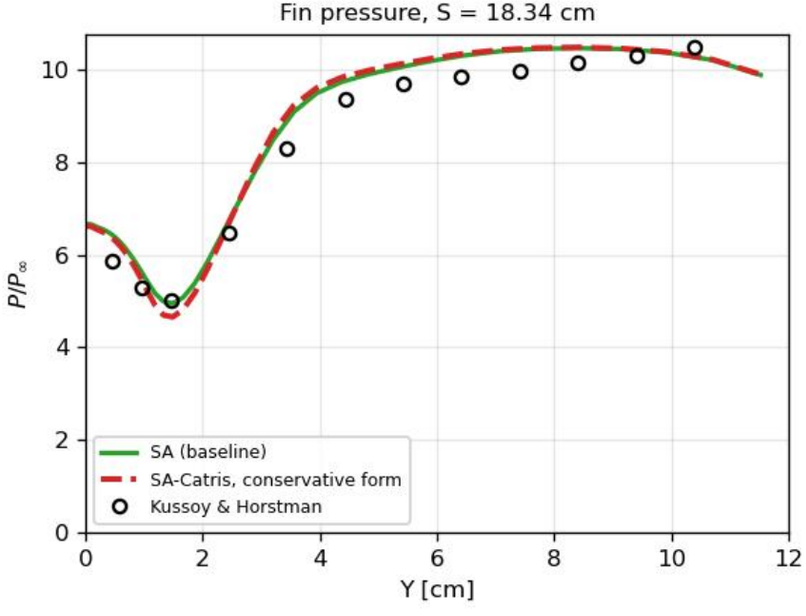


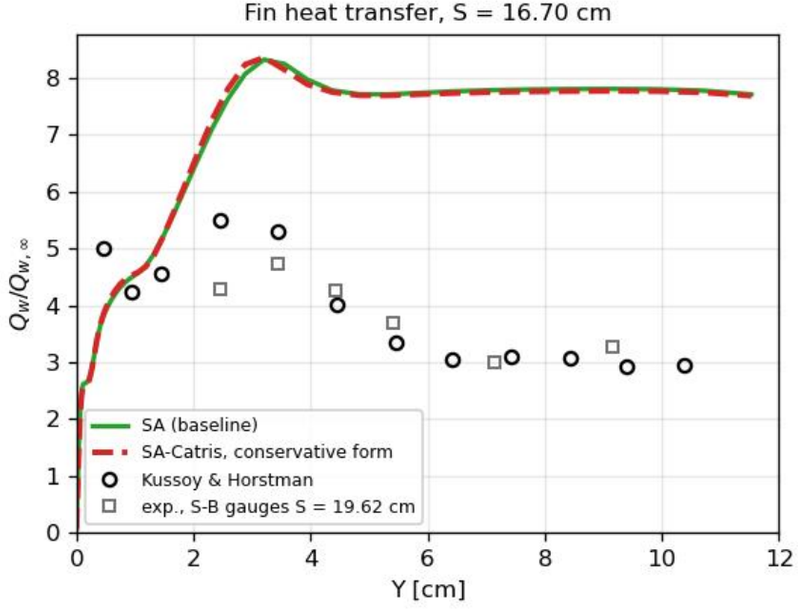
Kussoy & Horstman Mach 8.2, 15 deg sharp fin: SA-Catris, conservative form vs SA vs experiment
Catris-Aupoix density correction, conservative left-hand side (TMR SA-noft2-CatrisCons)



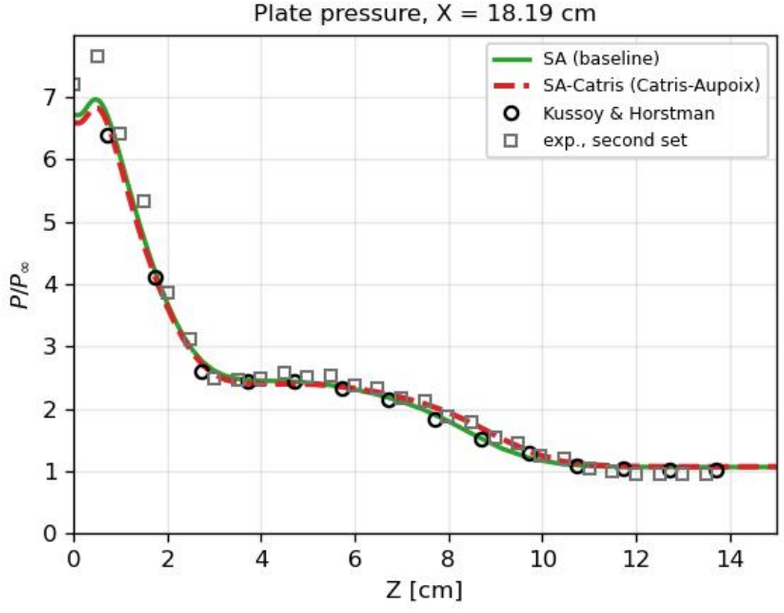


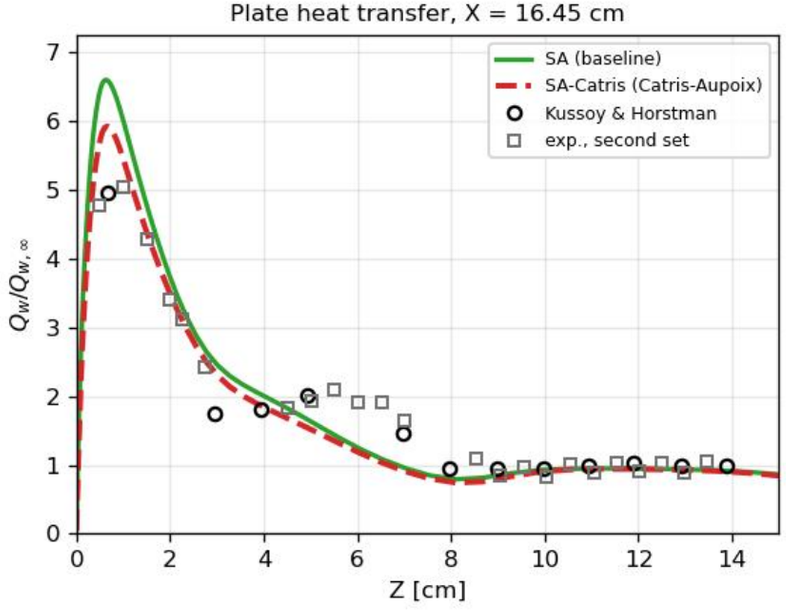


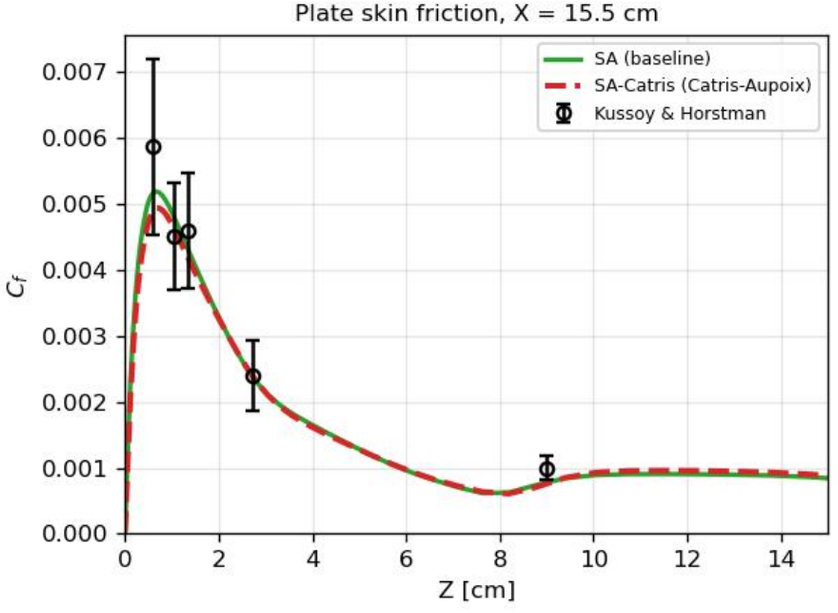


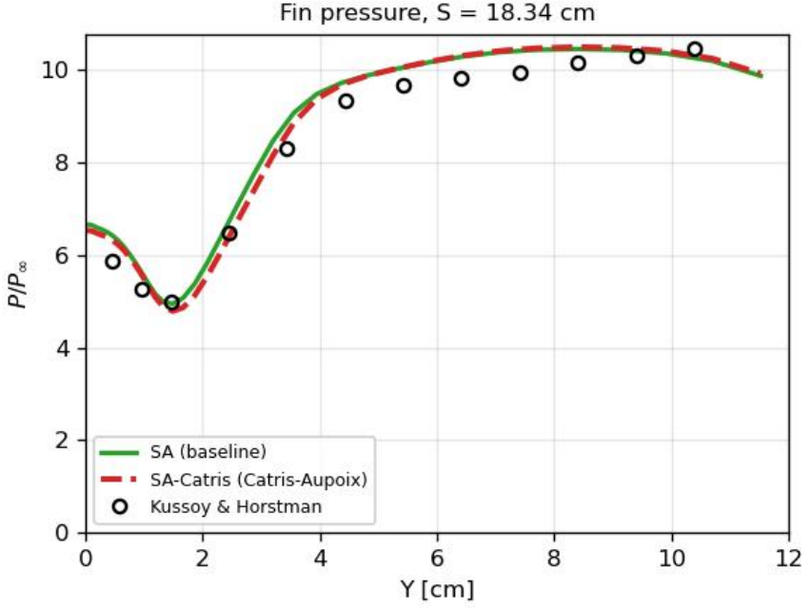


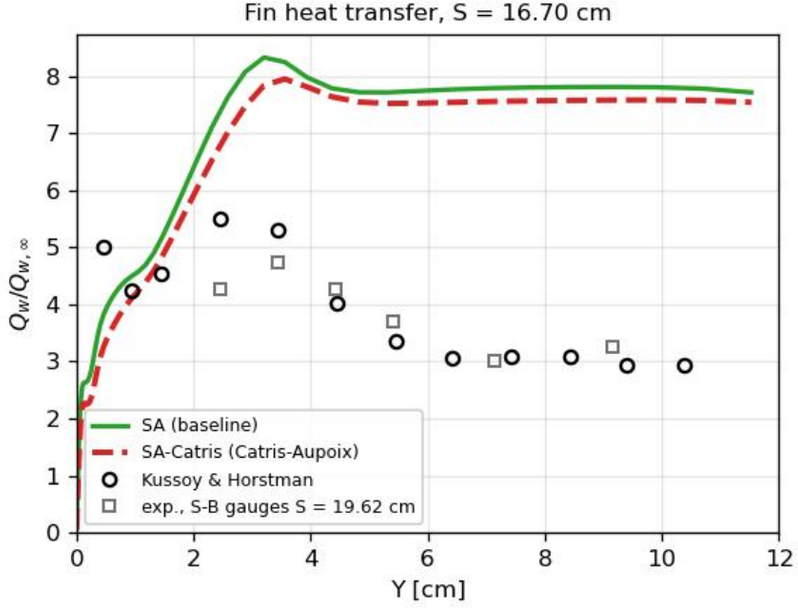
Kussoy & Horstman Mach 8.2, 15 deg sharp fin: SA-Catris (Catris-Aupoix) vs SA vs experiment
Catris-Aupoix density correction (2000): diffuses sqrt(rho) nuTilde, laminar diffusion of mu, original non-conservative left-hand side



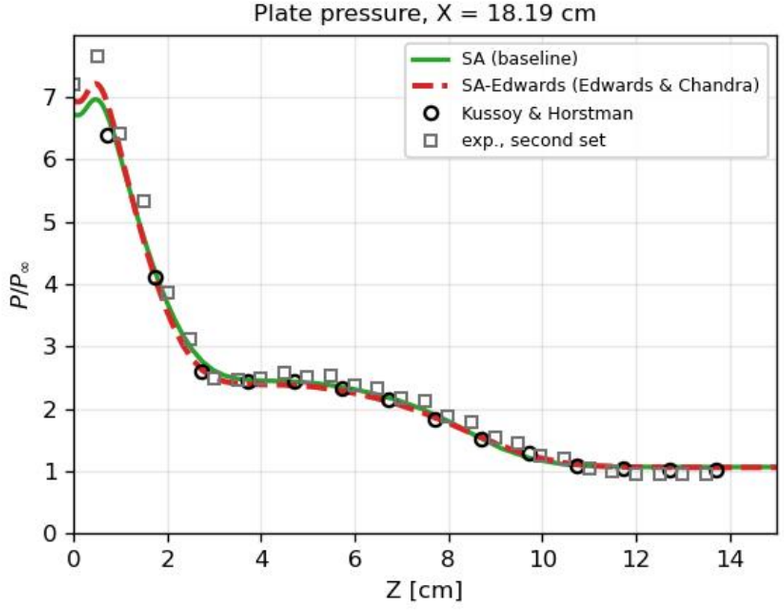


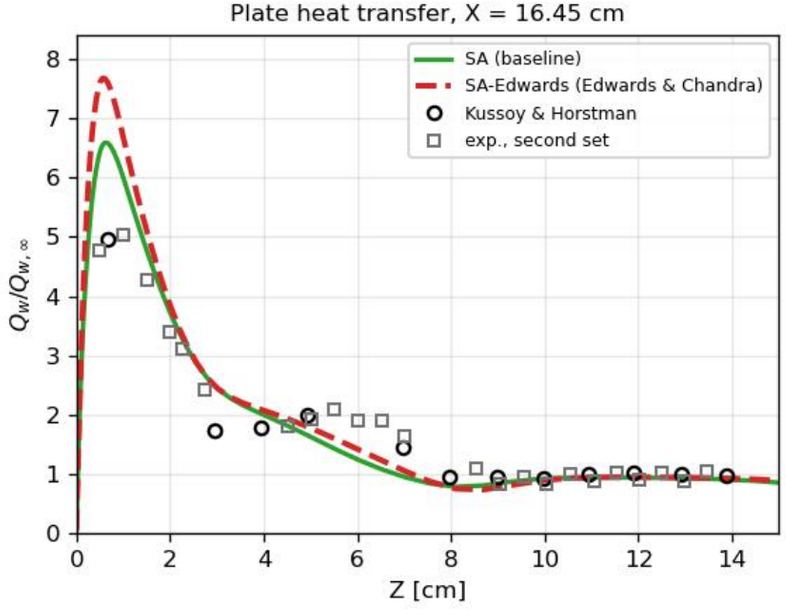


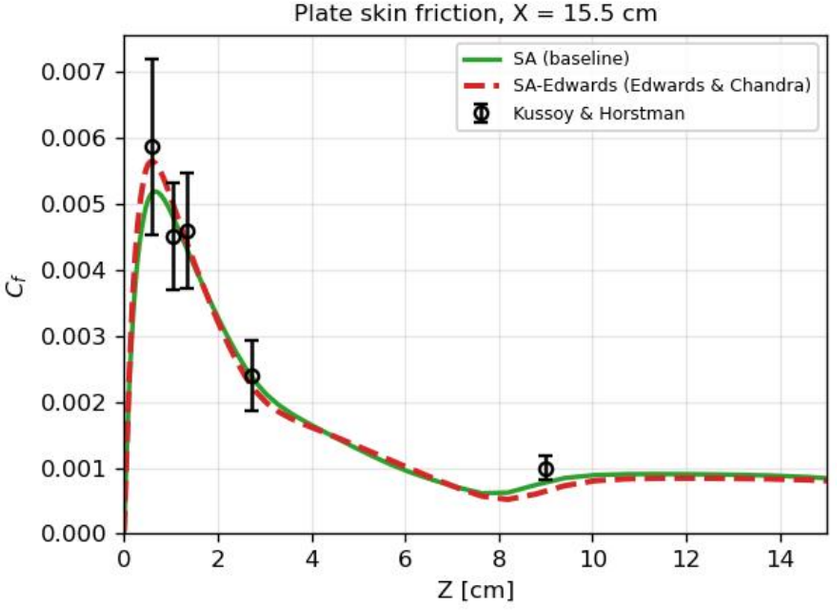


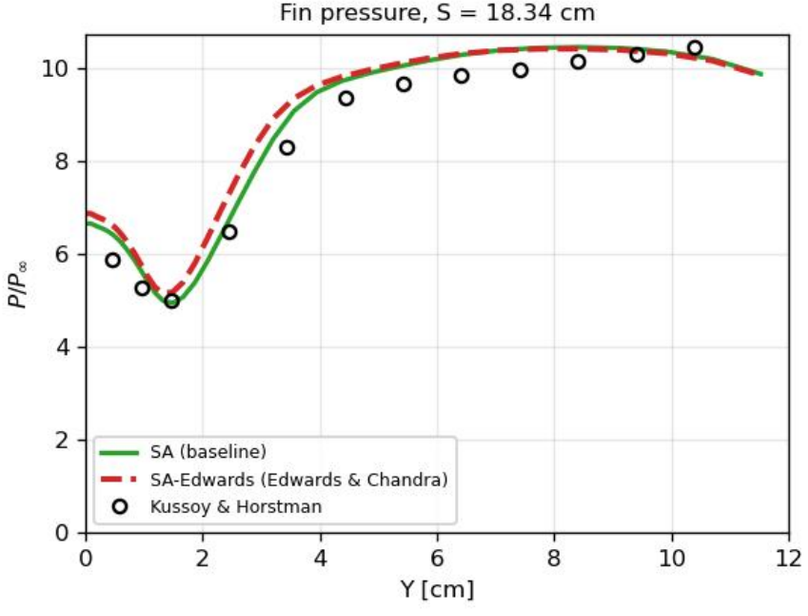


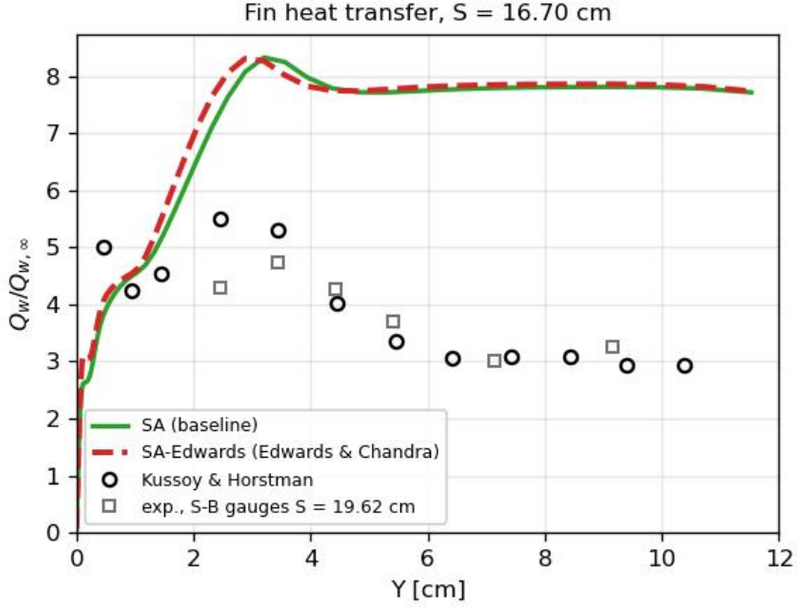
Kussoy & Horstman Mach 8.2, 15 deg sharp fin: SA-Edwards (Edwards & Chandra) vs SA vs experiment
Edwards & Chandra (AIAA 94-2275): strain-based production, Shat = S (1/chi + fv1), r = tanh(nuTilde/(Shat k^2 d^2))/tanh(1), no ft2



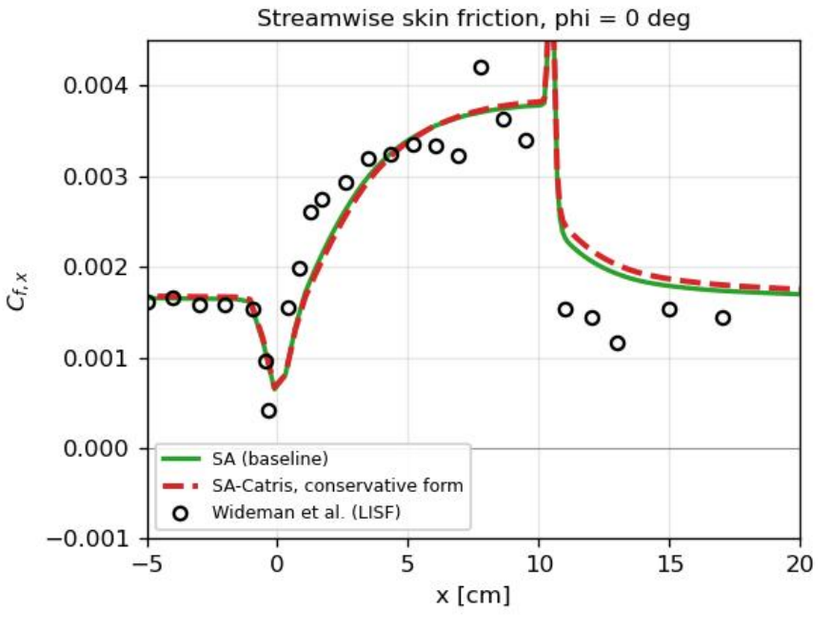


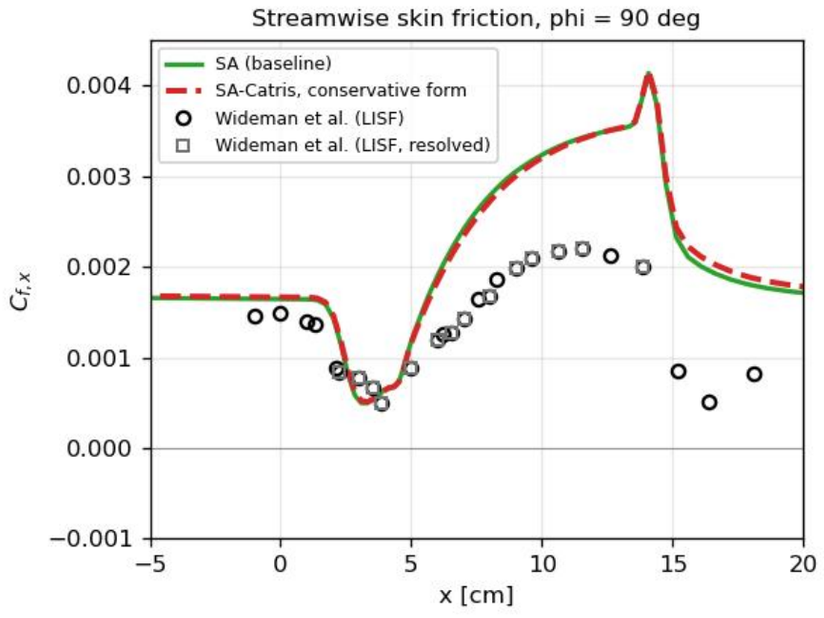


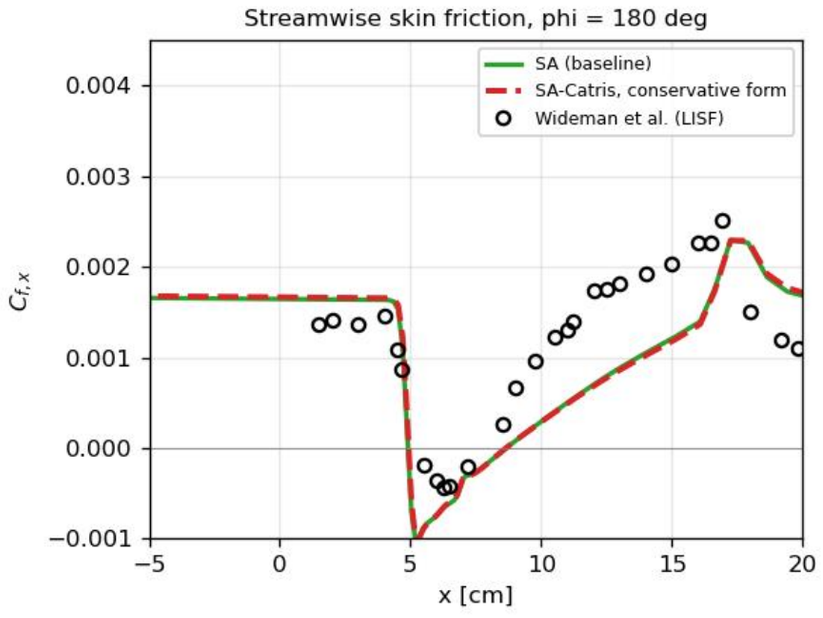


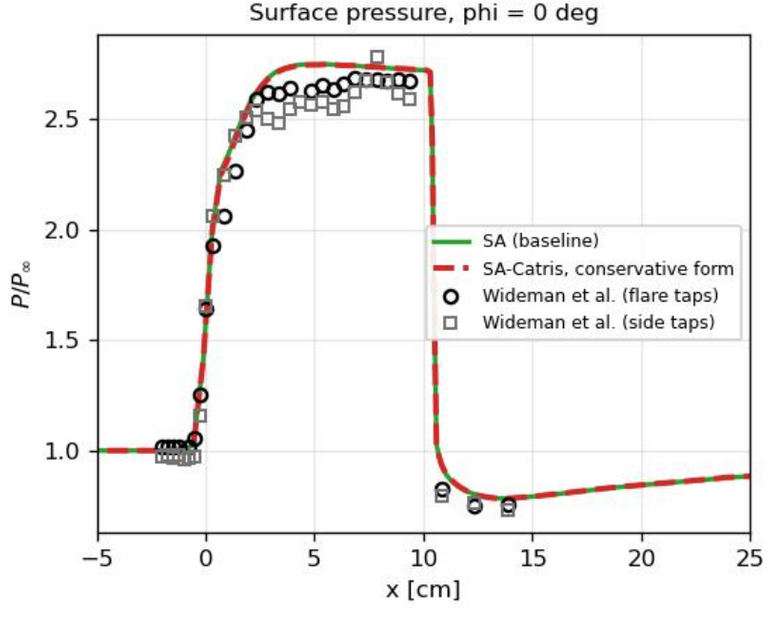


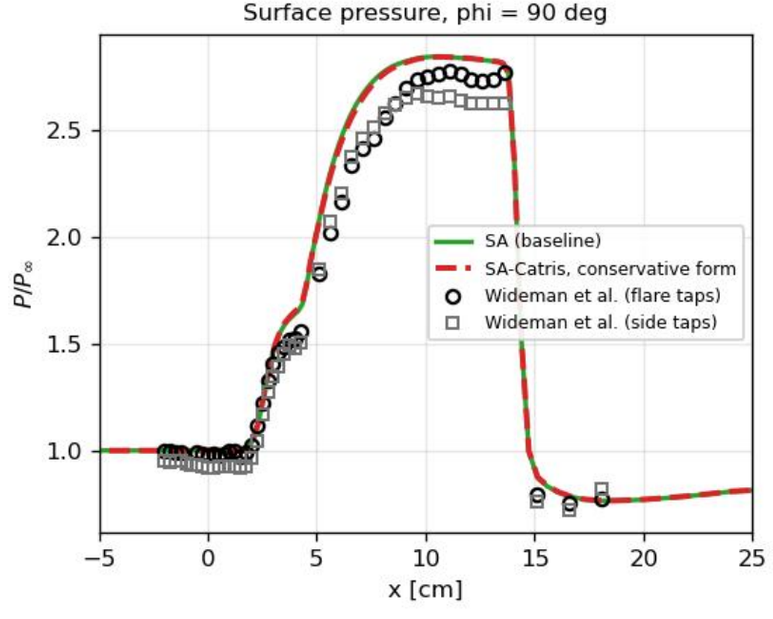
Wideman et al. Mach 2.89 cylinder / offset 20 deg flare: SA-Catris, conservative form vs SA vs experiment
Catris-Aupoix density correction, conservative left-hand side (TMR SA-noft2-CatrisCons)

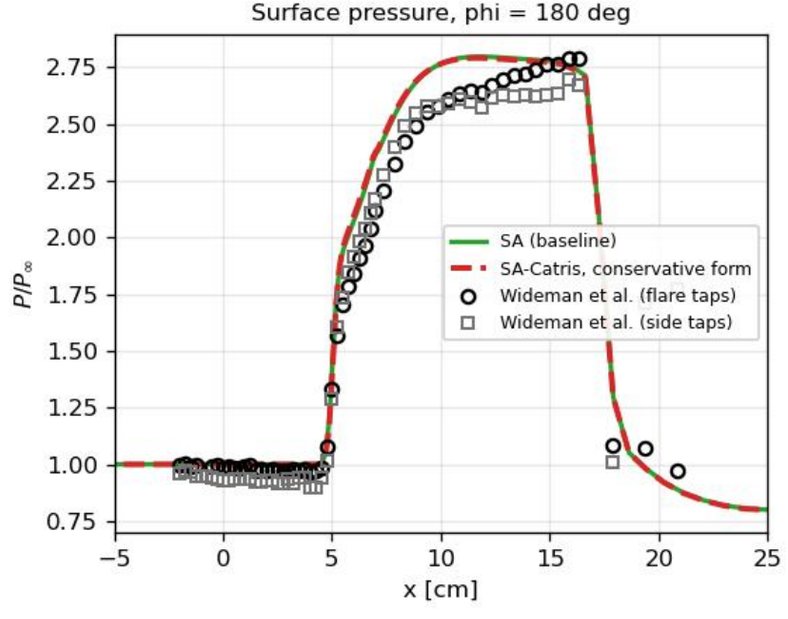




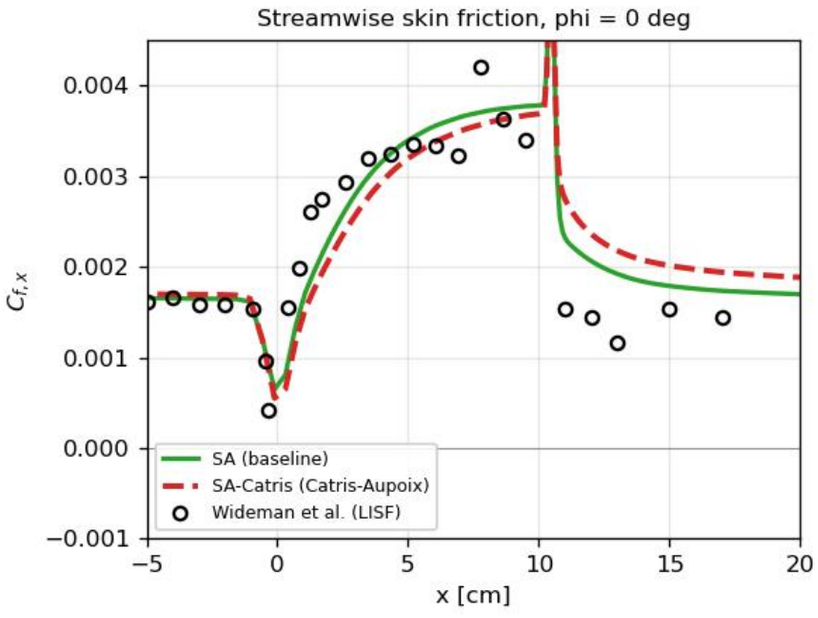


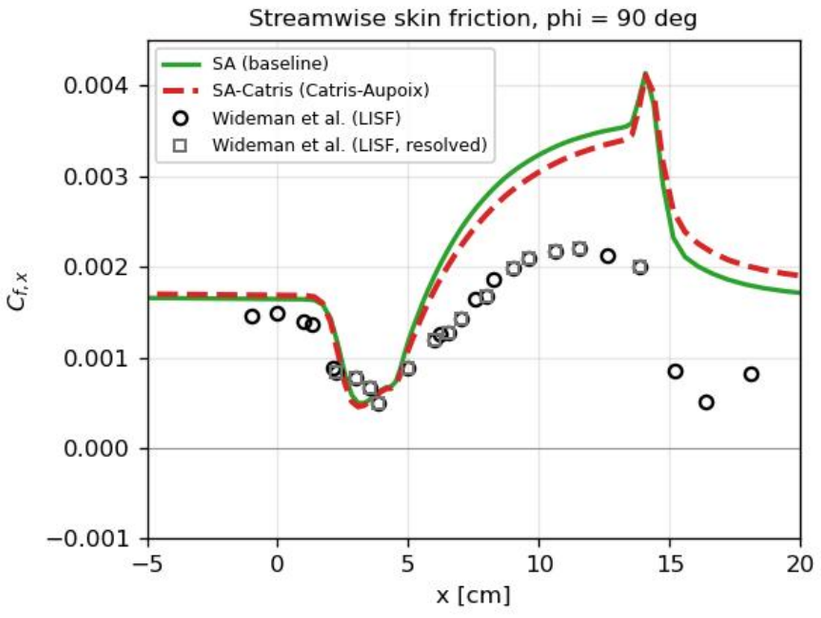


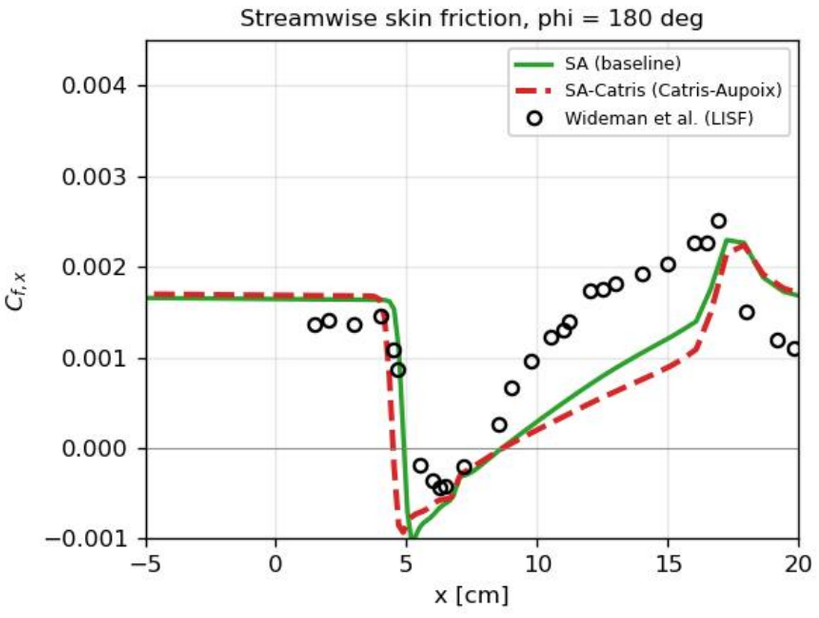


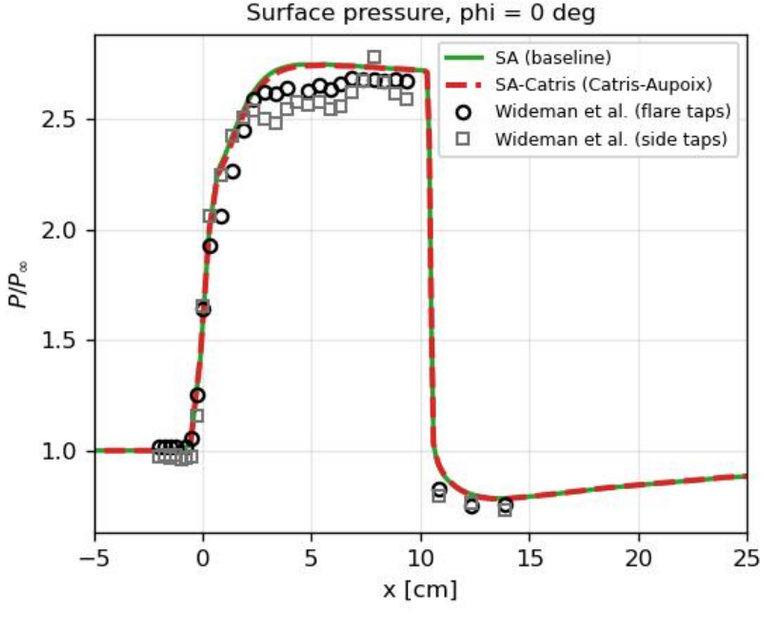


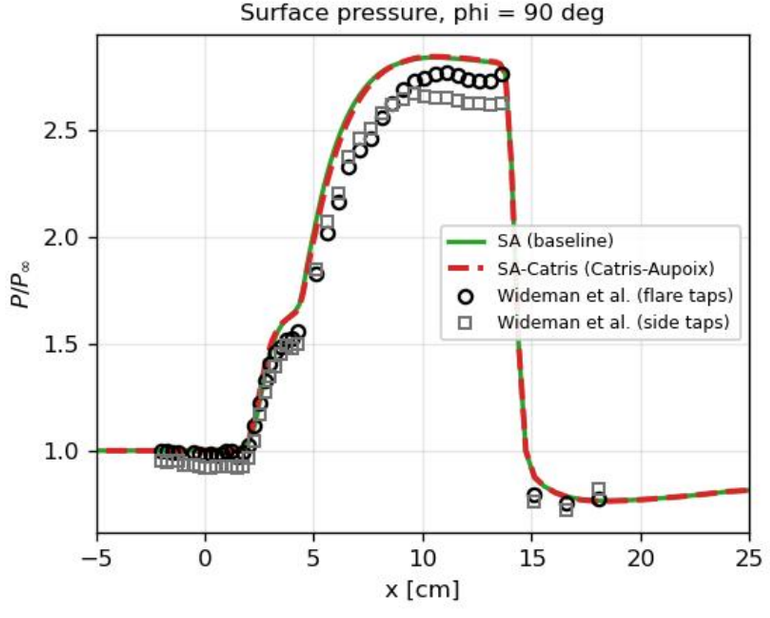
Wideman et al. Mach 2.89 cylinder / offset 20 deg flare: SA-Catris (Catris-Aupoix) vs SA vs experiment
Catris-Aupoix density correction (2000): diffuses sqrt(rho) nuTilde, laminar diffusion of mu, original non-conservative left-hand side

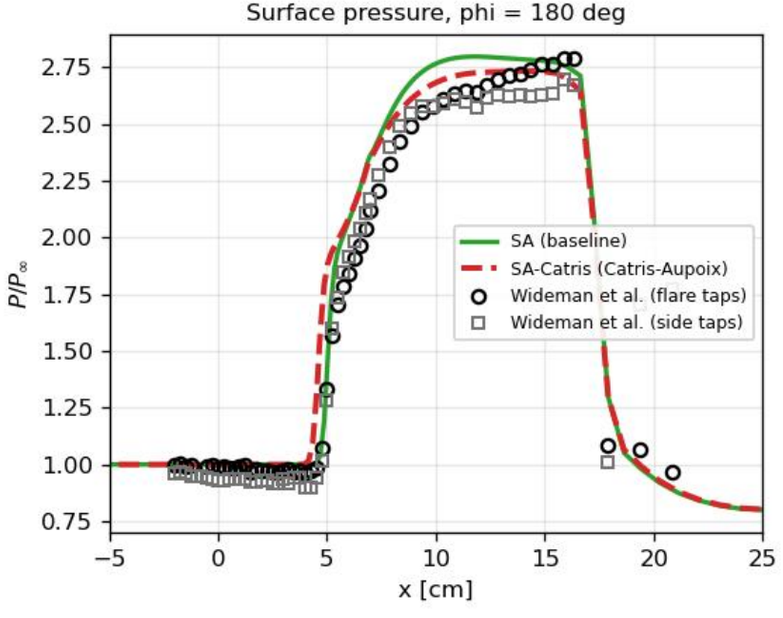




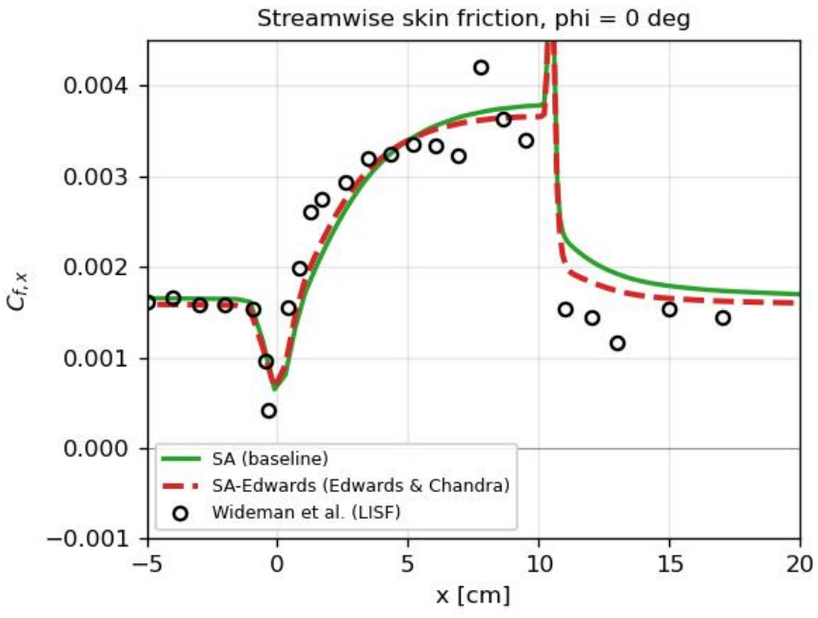


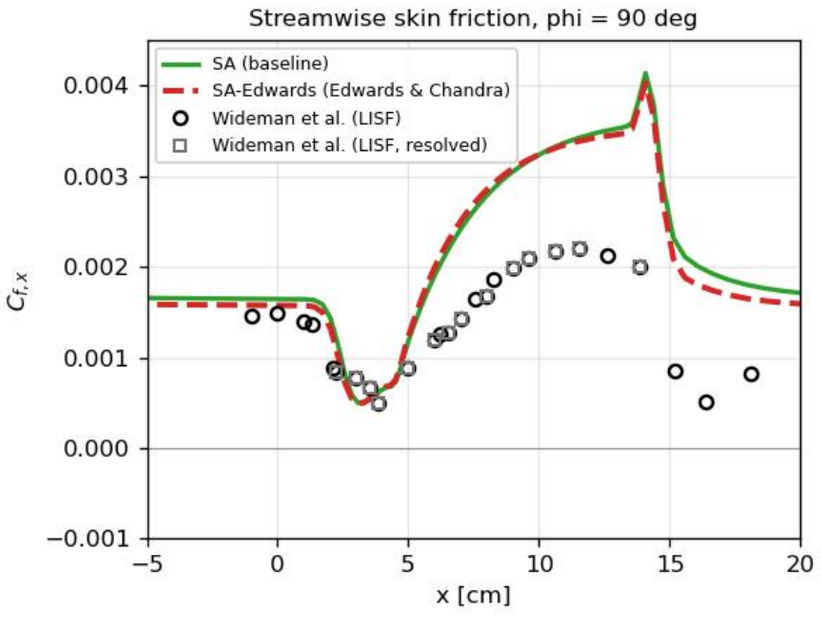


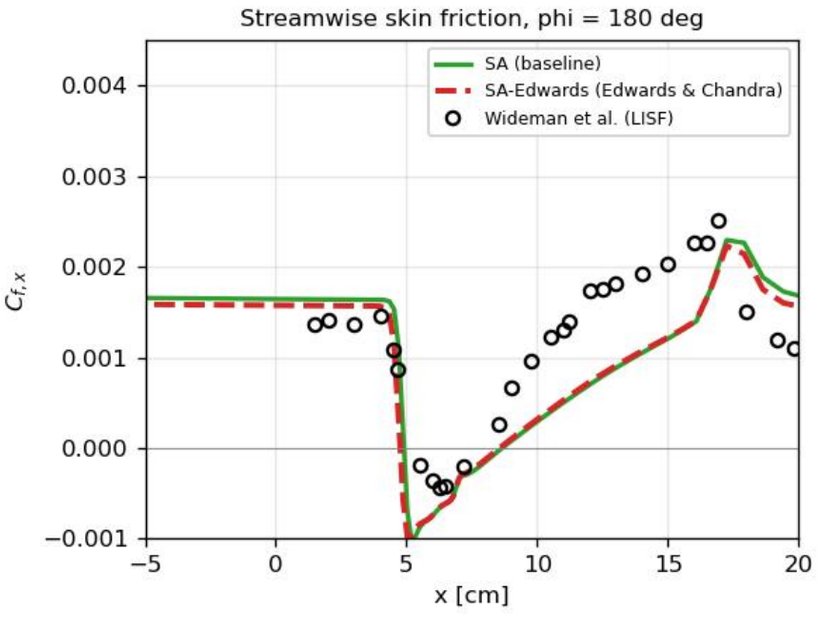


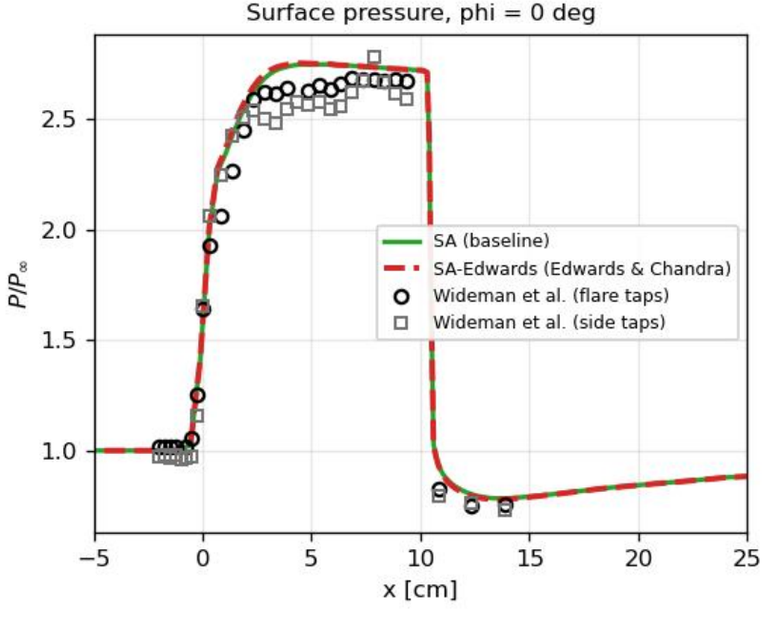


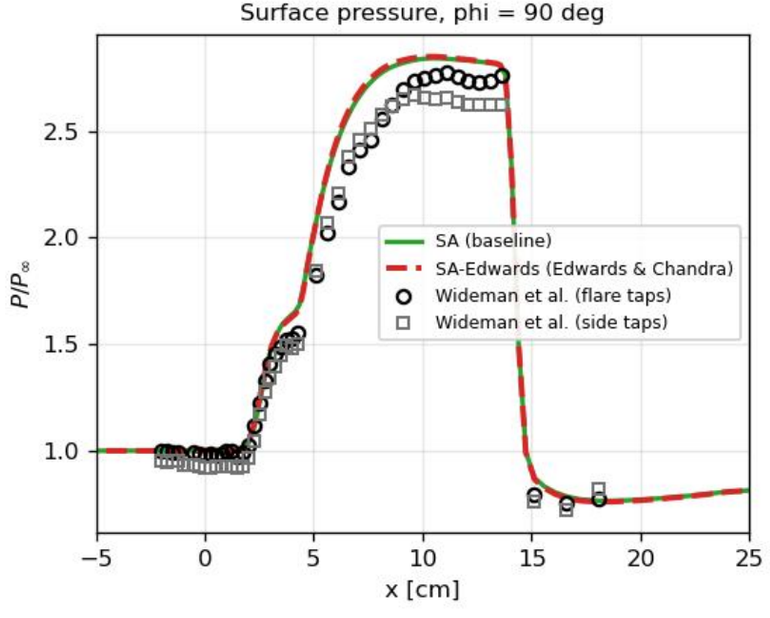
Wideman et al. Mach 2.89 cylinder / offset 20 deg flare: SA-Edwards (Edwards & Chandra) vs SA vs experiment
Edwards & Chandra (AIAA 94-2275): strain-based production, Shat = S (1/chi + fv1), r = tanh(nuTilde/(Shat k^2 d^2))/tanh(1), no ft2

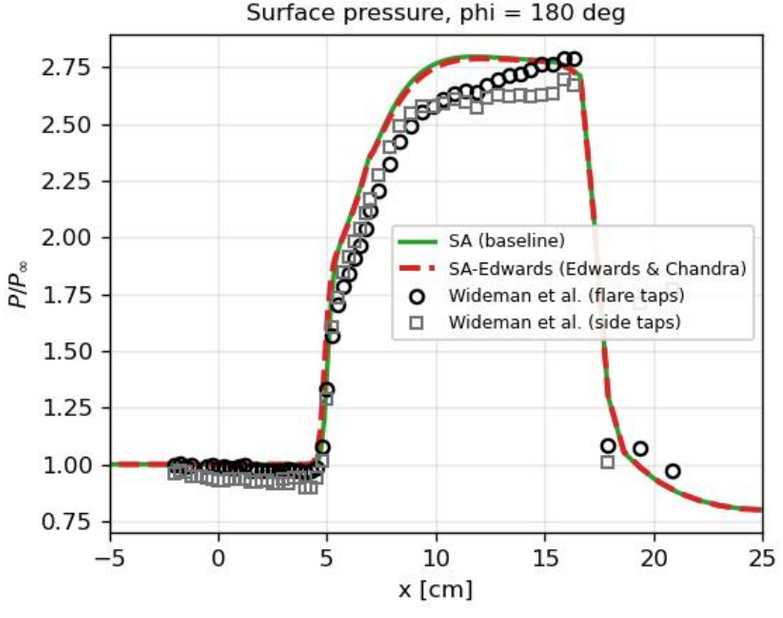




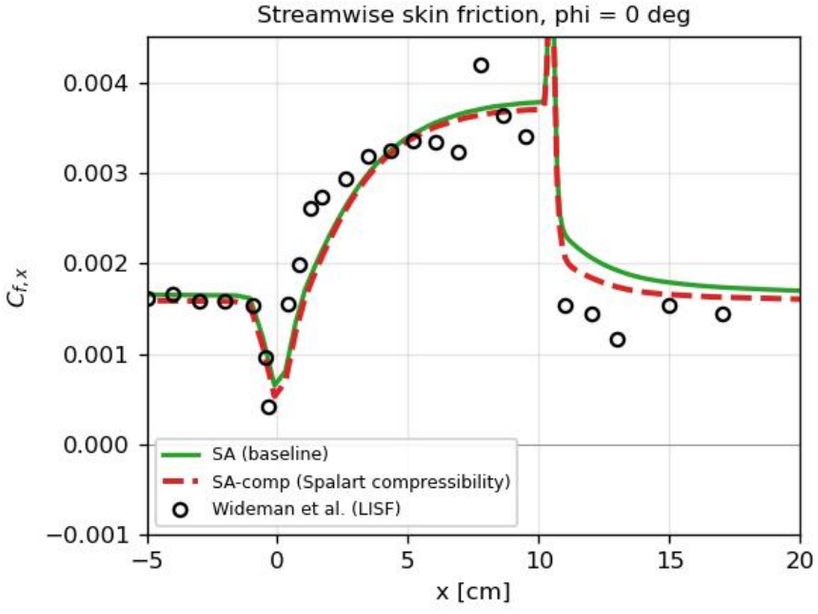


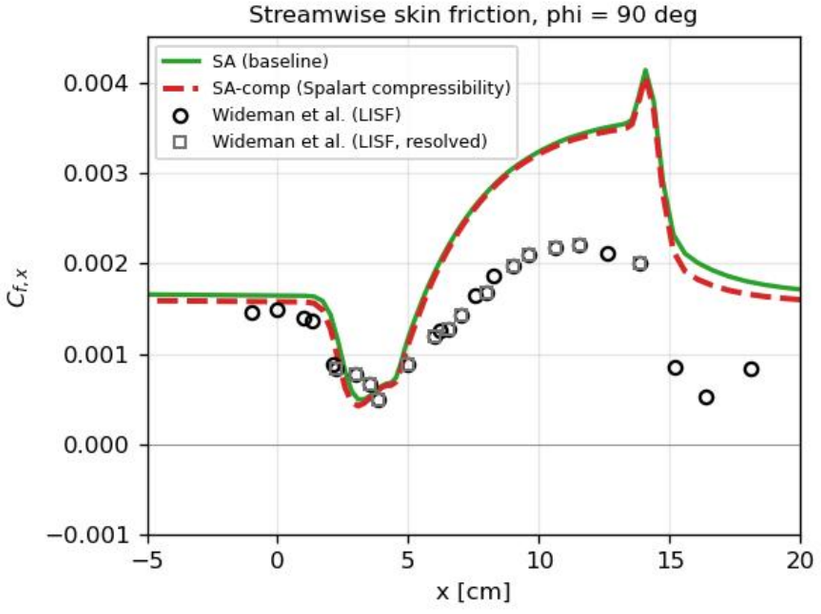


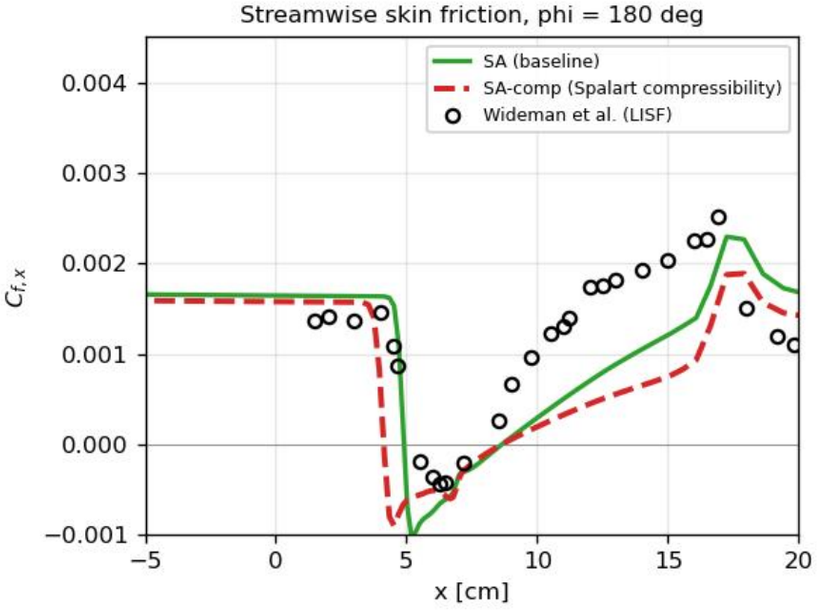


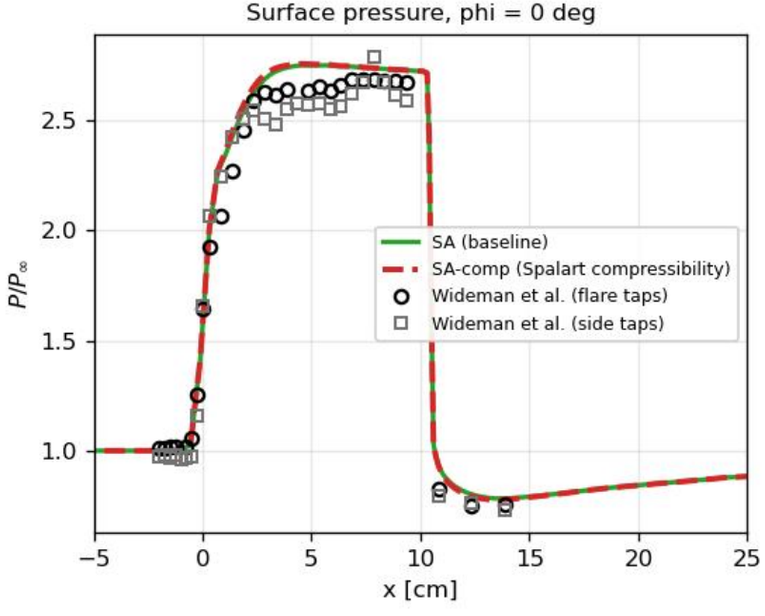


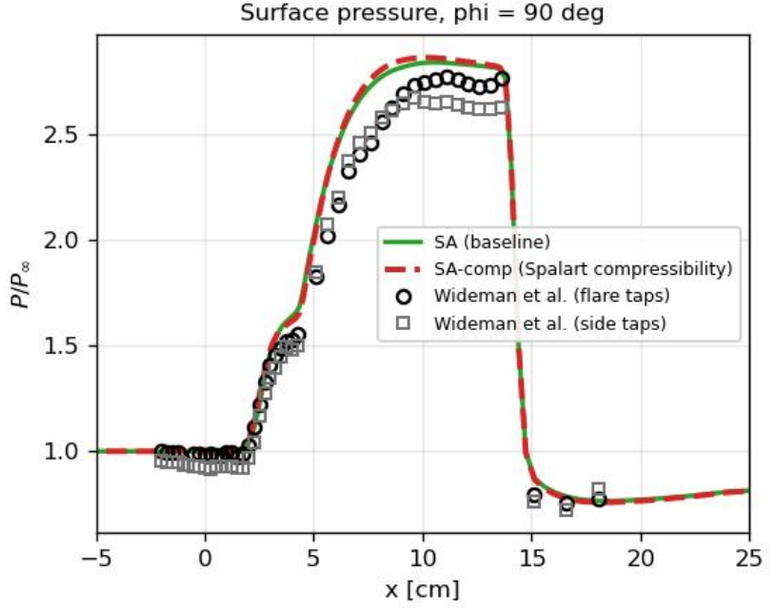
Wideman et al. Mach 2.89 cylinder / offset 20 deg flare: SA-comp (Spalart compressibility) vs SA vs experiment
Spalart compressibility correction (AIAA 2000-2306): extra destruction -C5 nuTilde^2 (du_i/dx_j)^2 / a^2, C5 = 3.5

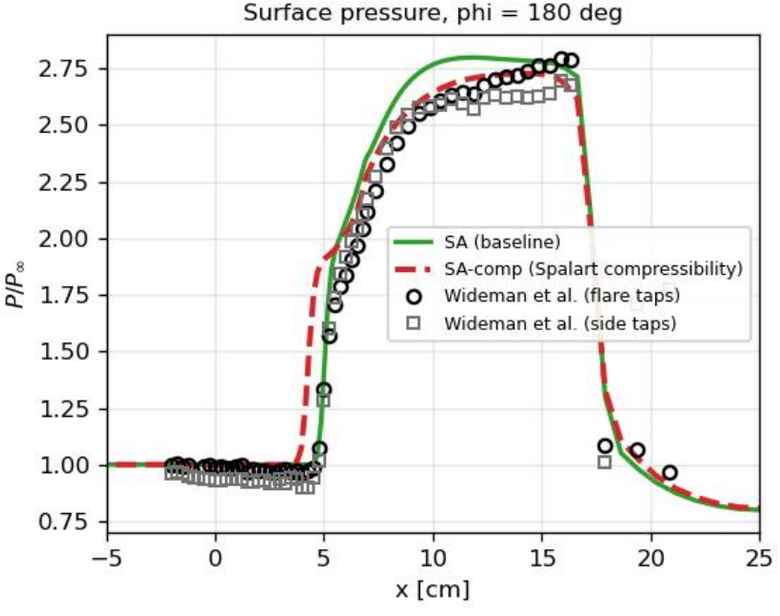


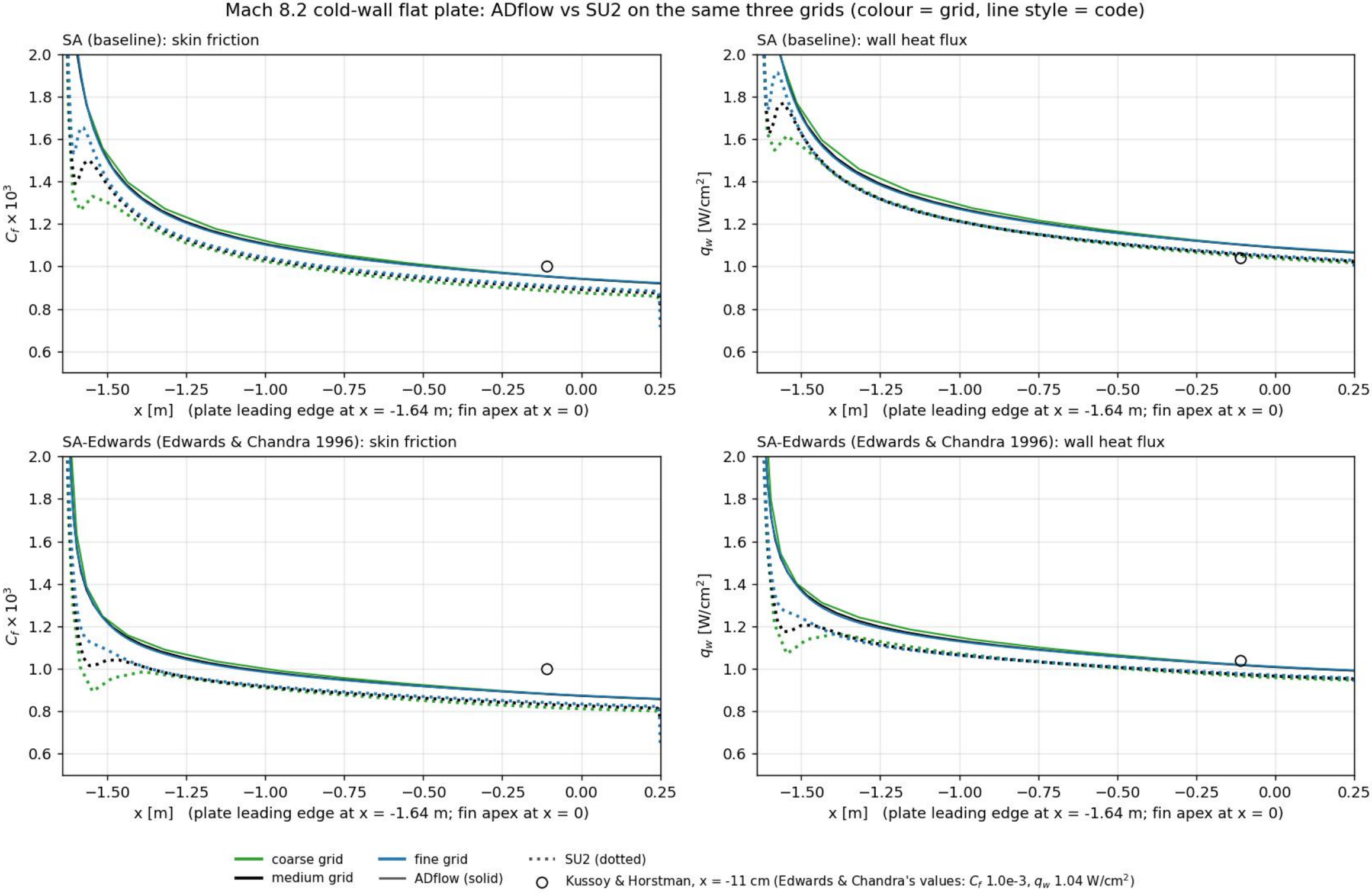


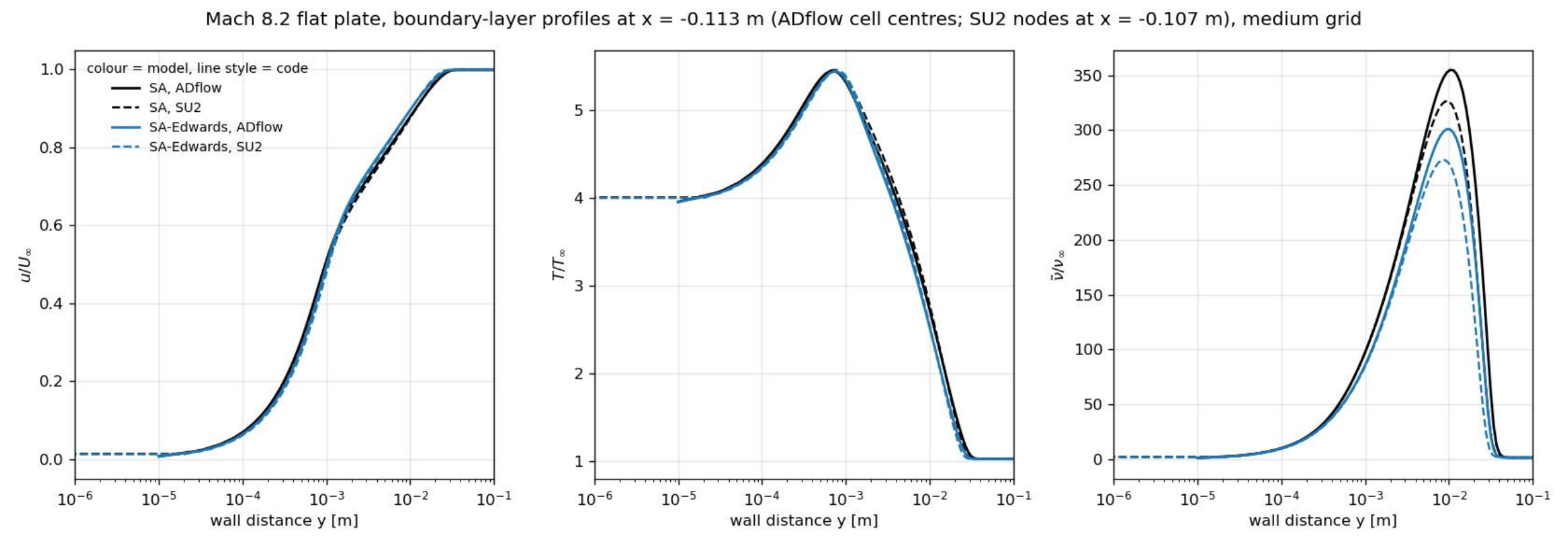









Mach 8.2 cold-wall flat plate: ADflow vs SU2 on the same three grids (colour = grid, line style = code)
SA (baseline): skin friction
SA (baseline): wall heat flux
SA-Edwards (Edwards & Chandra 1996): skin friction
SA-Edwards (Edwards & Chandra 1996): wall heat flux
$C_f \times 10^3$
q_w [W/cm^2]
x [m] (plate leading edge at x = -1.64 m; fin apex at x = 0)
coarse grid
medium grid
fine grid
ADflow (solid)
SU2 (dotted)
Kussoy & Horstman, x = -11 cm (Edwards & Chandra's values: C_f 1.0e-3, q_w 1.04 W/cm^2)


Mach 8.2 flat plate, boundary-layer profiles at x = -0.113 m (ADflow cell centres; SU2 nodes at x = -0.107 m), medium grid
colour = model, line style = code
SA, ADflow
SA, SU2
SA-Edwards, ADflow
SA-Edwards, SU2
u/U∞
T/T∞
ν̃/ν∞
wall distance y [m]
wall distance y [m]
wall distance y [m]